

Relativistic corrections in satellite positioning systems



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1. Introduction

Satellite geolocation systems, or GNSS (Global Navigation Satellite System), are introduced to determine the position of any observer on the Earth's surface and are based on a localization principle known as trilateration, which consists of measuring different distances to points whose locations are already known.

If a constellation of satellites in orbit emits signals at a given speed, any observer on the ground can determine the distance separating them from each satellite by measuring the time it takes for the signal to arrive. Thus, through the trilateration resulting from several distances, the observer can determine the specific point at which they are located with respect to the satellites, and by knowing the satellites' positions, they can also determine their own position on the surface.

To achieve this, a measurement of that time is required, and in the case of perfect time synchronization, it would be enough to send a single synchronous timestamp with all signals so that the observer could compare the received times with their own and obtain those distances.

However, satellites move at high velocities and altitudes above the Earth's surface, making relativistic effects significant, and these effects are further amplified by the propagation speed of the signal, which is the speed of light in vacuum.

Under these conditions, if no correction were introduced for the measurement of proper time on board the satellites, synchronization would quickly be lost after any resynchronization event, making it impossible to establish those distances reliably and to determine position accurately from them.

It is important to highlight that the relativistic correction is not the only one required in satellite geolocation, because there are other effects that introduce measurement errors on the ground—for example, those derived from the rotation of the observer comoving with the Earth and affected by its rotation (Sagnac effect), or those caused by the chemical composition of the atmosphere, which delays the signals.

The purpose of this document is to detail the relativistic corrections required due to altitude and velocity. It also describes the basic principles of trilateration for satellite positioning and includes several demonstrations and notes on analytical mechanics that may be useful in the context of general relativity and satellite navigation.

All images included in this document are either created by the author or generated with Google's Nano Banana 2. For all numerical approximations and other undeveloped expressions common to both special and general relativity, the following sources have been required

Sources	
#1	Spacetime and Geometry: An Introduction to General Relativity , Carroll, S. M.
#2	Relativity in the Global Positioning System , Ashby N.
#3	Relativistic Corrections in the European GNSS Galileo , ESA, Mudrak, A., De Simone, P., Lisi, M. (2015)
#4	Geodetic Reference System 1980 , IUGG
#5	Solar System Dynamics , JPL
#6	GPS Space Segment (ICD) , U.S. Department of Defense
#7	GLONASS Interface Control Document (ICD) , UNAVCO
#8	Galileo Interface Control Document (ICD) , European GNSS Service Centre
#9	BeiDou Interface Control Document (ICD) , China Satellite Navigation Office (CSNO)

2. Trilateration

2.1. Trilateration in the plane

Let us imagine a plane and a circle C_1 whose points are all at a distance r_1 from its center o_1 . No observer on the circumference can determine their exact position by measuring the radius r_1 —only that they lie somewhere on the circle. Now imagine another circle with radius r_2 and center o_2 at an arbitrary distance from o_1 . Both circles have two intersection points, p_1 and p_2 (figure 1).

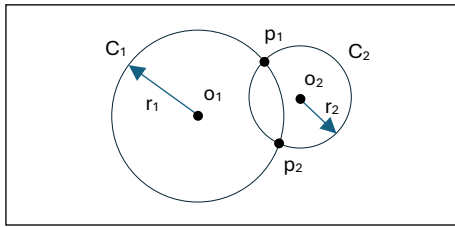


Figure 1

We can see that by knowing r_1 and r_2 , the observer can determine that they are either at p_1 or p_2 with respect to o_1 and o_2 , but not which of the two. We can also see from the very definition of the problem that there are infinitely many circles passing through those two points, as many as can be drawn with centers on the line defined by the centers of the two circles, since that line is the perpendicular bisector of the segment joining p_1 and p_2 (figure 2).

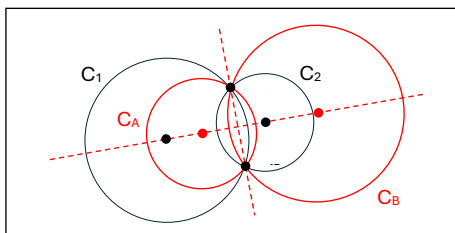


Figure 2

Now introduce a third circle C_3 with the only condition that its center not lie on the line joining the previous two centers (figure 3).

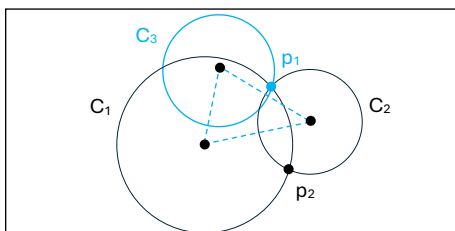


Figure 3

We see that there is no circle that, while passing through one of the intersection points of two circles, can also pass through the other: point p_1 is thus determined by the three radii of circles C_1 , C_2 , and C_3 . Determining a position in the plane using this

procedure from three distances to points of known position is called trilateration.

2.2. Trilateration in space

Let us now try to extend this process to our three-dimensional space. Observe that the equivalent of a circle in the plane is the surface of a sphere S_1 , where all points are at a distance r_1 from its center o_1 . No observer on the surface of the sphere can determine their exact position—only that they lie somewhere on that surface. Now imagine another sphere S_2 with radius r_2 and center o_2 at an arbitrary distance from o_1 . The surfaces of both spheres share an entire circle as their intersection, C_{12} (figure 4).

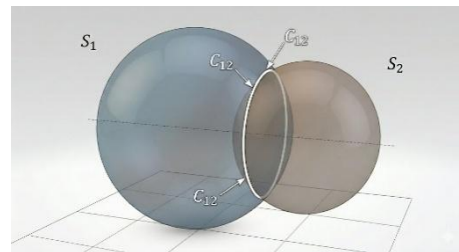


Figure 4

Again, from the definition of the problem, there are infinitely many spheres whose surfaces share the intersection C_{12} , provided their centers lie on the line joining the two original centers, since that line is the axis perpendicular to the plane of the intersection circle C_{12} passing through its center (figure 5).

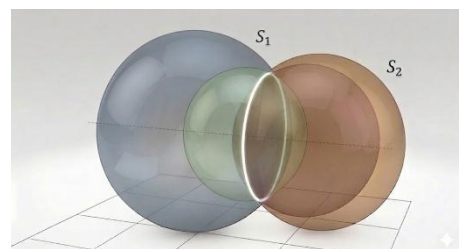


Figure 5

Now introduce a third sphere S_3 with the only condition that its center not lie on the line joining the centers of the previous two. The intersection of the surfaces of the three spheres is now reduced to two points, which are the intersection of the surface of S_3 with

the circle resulting from the intersection of the other two, C_{12} (figure 6).

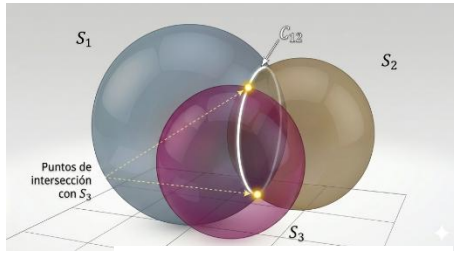


Figure 6

However, both points are equally possible, making a fourth sphere S_4 necessary. That fourth sphere cannot lie in the same plane as the centers of the other three, or we would again face a similar indeterminacy, finding infinitely many spheres whose surfaces intersect at those two points. We see that it is impossible to find a fourth sphere whose center does not lie in that plane and whose surface passes through both intersection points of the previous three surfaces, so point p_1 becomes fully determined (figure 7).

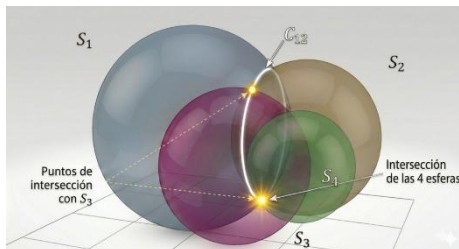


Figure 7

This is the positioning system in space used by GNSS systems.

2.3. Satellite trilateration

Here it is worth noting that when this procedure is used to determine satellite-based geolocation, the Earth itself resembles a sphere, so it would be sufficient to use three additional spheres (three satellites) to determine the position. An observer on the Earth's surface would know which of the two possible intersection points of the other three sphere surfaces corresponds to their location, because the other intersection point would be inaccessible, thus determining their position (figure 8).

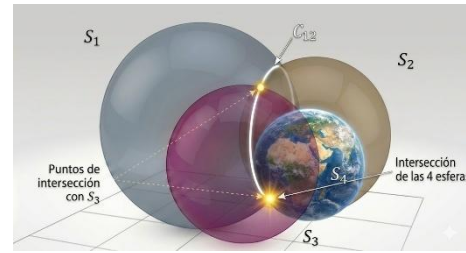


Figure 8

However, at least four satellites are used for reasons related to the ability to measure the observer's proper time on the ground, which lacks the required precision and synchronization.

We know how to trilaterate a position in space using four fixed points, but how do we solve trilateration when the centers of the spheres are moving? This is the case for satellites, whose velocities are significant and which are no longer at the same position relative to the receiver on the surface by the time their signals are received.

Thus, satellites must transmit not only a timestamp but also their position at the moment of emission. Once other corrections are applied, the geolocation problem becomes one of determining the position of points on the surface relative to the satellites.

Satellites therefore act as position references, self-determined from the stability of their orbits, and broadcast in the form of ephemerides, i.e., orbital parameters describing their trajectory at each moment. They also require corrections for any perturbations in their orbital mechanics. To ensure the accuracy of all measurements—both position and time—periodic updates are sent to the satellites from the ground.

For this reason, [medium Earth orbits](#) (MEO), non-geostationary and with periods shorter than one day, are used. These allow frequent updates from the ground while keeping satellites in free fall, as will be detailed later. As a curiosity, satellites also broadcast the constellation's almanac, i.e., not only their own ephemerides but also those of the other satellites, helping the ground receiver identify and validate signals from other satellites at any given moment.

3. Relativistic Corrections

We now proceed to specify the two relativistic corrections that must be implemented at the level of the satellites' proper-time clocks. Among all required corrections, these two are the only ones that are common to all observers on Earth's surface, regardless of their location. For this reason, they are the only corrections configured directly into the satellites' clock offsets before launch.

The remaining corrections do not apply uniformly across the surface and are therefore handled in software at the receiver level. For example, the correction required due to Earth's rotation (Sagnac effect) depends on the observer's latitude: surface velocity is maximal at the equator and zero at the poles. Similarly, atmospheric delays depend on the chemical composition of the atmosphere, which is not homogeneous across the globe.

Furthermore, among all necessary corrections, the non-relativistic ones (Sagnac, atmospheric composition, etc.) represent only a small fraction of the total effect—although they cannot be neglected if precise satellite navigation is desired. The two relativistic effects account for almost the entire correction, with the gravitational term being the dominant contribution.

3.1. Imprecision in the Observer's Time Measurement on the Ground

All signals travel at the speed of light in vacuum ($\sim 10^8$ m/s), meaning that microscopic timing errors translate into unacceptable distance errors. If we compute the distance to satellites using

$$(3.1) \quad d = c \cdot \Delta t$$

we see that each $1 \mu\text{s}$ of error ($\sim 10^{-6}$ s) results in a distance error of hundreds of meters ($\sim 10^2$ m), and this is for a single satellite. Such errors would accumulate with relativistic effects until corrected again from the ground, severely limiting the system's precision and usefulness.

For this reason, satellites are equipped with highly precise proper-time measurement technologies such as atomic clocks. However, ground receivers cannot be

equipped with such technology. This is why we do not use only three satellites to determine surface position, but four.

Using a fourth satellite allows the receiver to compute position accurately without depending on the observer's own imperfect time measurement. Moreover, it enables portable devices—without precise clocks—to benefit from the timing precision of the satellites' atomic clocks. In practice, even more satellites are used to mitigate possible defects or reception errors (e.g., multipath reflections in urban environments).

Ground observers compute distances to satellites using the equations:

$$(3.2) \quad (x - x_i)^2 + (y - y_i)^2 + (z - z_i)^2 = d_i^2 = (c \cdot \Delta t_i)^2 = c^2 \Delta t_i^2$$

where x_i is the position of satellite i , d_i the distance, and x the observer's position. Let b be the difference between the observer's proper time and the satellites' time (assuming perfect synchronization among satellites). Then:

$$(3.3) \quad (x - x_i)^2 + (y - y_i)^2 + (z - z_i)^2 = d_i^2 = c^2 \cdot (t_i - t)^2 = c^2 b^2$$

and therefore:

$$(3.4) \quad \sqrt{(x - x_i)^2 + (y - y_i)^2 + (z - z_i)^2} - c \cdot b = 0$$

We can write one such equation for each satellite. The sign of the time difference does not matter, since it is absorbed by the squared distance.

Thus, we have a system of four equations with four unknowns, one of which (the clock bias b) is common to all. Four satellites allow us not only to determine the position but also to solve for b , enabling the receiver to adjust its own time and effectively inherit the precision of the satellites' atomic clocks.

3.2. Relativistic Corrections Due to Velocity

We know that the proper time of two observers in motion flows differently depending on their relative velocities. Ideally, we would compute the velocities of both satellites and ground observers in the same reference frame. However, surface velocity is

not uniform, so we adopt a reference frame centered at Earth's center and non-rotating with it: the Earth-Centered Inertial (ECI) frame.

In this frame, ground observers are nearly static compared to satellites, and software corrections are applied at the receiver level. As we will see numerically, the ratio between orbital velocities and surface velocities is indeed large, validating this approach for special-relativistic treatment in satellite navigation.

3.2.1. Free-Fall Velocity (Circular Orbit)

For Earth's gravitational field, we use the [Schwarzschild](#) metric, treating Earth as a perfect radial symmetry:

$$(3.5) \quad ds^2 = -c^2 \left(1 - \frac{r_s}{r}\right) dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + r^2 d\Omega^2 \quad \text{where } r_s = \frac{2GM}{c^2}$$

This solution imposes an asymptotically flat spacetime, which is already weak under satellite operating conditions, where Newtonian mechanics would suffice.

By spherical symmetry, we restrict motion to the equatorial plane ($\theta = \pi/2$, $\dot{\theta} = 0$). The spatial velocity components in this plane are the radial velocity v_r and the tangential velocity v_l (associated with the angular coordinate ϕ). Without symmetry, two angular components would appear.

This is consistent with the Christoffel symbols under these conditions:

$$(3.6) \quad \begin{aligned} \Gamma_{\theta\theta}^\phi &= 0 \\ \Gamma_{\theta\phi}^\phi &= \Gamma_{\phi\theta}^\phi = \cot \theta = \cot \pi/2 = 0 \end{aligned}$$

Using the [radial geodesic](#) equation:

$$(3.7) \quad \ddot{r} - \frac{r_s}{2r(r-r_s)} \dot{r}^2 - r \left(1 - \frac{r_s}{r}\right) (\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2) + c^2 \frac{r_s}{2r(r-r_s)} \dot{t}^2 = 0$$

and particularizing to the equatorial plane

(3.8)

$$\ddot{r} - \frac{r_s}{2r(r-r_s)} \dot{r}^2 - r \left(1 - \frac{r_s}{r}\right) \dot{\phi}^2 + c^2 \frac{r_s}{2r(r-r_s)} \dot{t}^2 = 0$$

we obtain the tangential velocity by imposing $\dot{r} = 0$ and dividing by $\dot{t}^2$:

(3.9)

$$\begin{aligned} -r \left(1 - \frac{r_s}{r}\right) \dot{\phi}^2 + c^2 \frac{r_s}{2r(r-r_s)} \dot{t}^2 &= 0 \\ \Rightarrow -r \left(1 - \frac{r_s}{r}\right) \frac{v_l^2}{r^2} + c^2 \frac{r_s}{2r(r-r_s)} &= 0 \end{aligned}$$

which yields:

(3.10)

$$v_l^2 = \frac{GM r}{(r-r_s)^2} \Rightarrow v_l = \frac{\sqrt{GM r}}{r-r_s}$$

which is the coordinate tangential velocity in radial orbits. In the calculation of time dilation we must use the [local velocity](#) but in the case of satellite operation we have that this expression is valid because the time dilation is not appreciable ($r \gg r_s$) $\Rightarrow d\tau \approx dt$. We can further simplify the above expression because and therefore use $(r - r_s) \approx r$

(3.11)

$$v_l = \frac{\sqrt{GM r}}{r} = \sqrt{\frac{GM}{r}}$$

This matches the Newtonian orbital velocity obtained by equating centripetal and gravitational forces:

(3.12)

$$F_g = F_c \Rightarrow G \frac{Mm}{r^2} = \frac{mv^2}{r} \Rightarrow v = \sqrt{\frac{GM}{r}}$$

3.2.2. Relativistic Correction in Free Fall (Circular Orbit)

We now compute the time dilation experienced by static observers relative to moving ones, using the Lorentz factor:

(3.13)

$$\gamma = \frac{dt}{d\tau} = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

which for $v \ll c$ becomes

(3.14)

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \approx \frac{1}{1 - \frac{1}{2} \frac{v^2}{c^2}}$$

Thus, the difference in proper-time flow is

(3.15)

$$\begin{aligned} \Delta\tau_{\text{velocity}} &= \frac{dt - d\tau}{dt} = 1 - \frac{d\tau}{dt} = 1 - \frac{1}{\gamma} \\ &= 1 - \left(1 - \frac{1}{2} \frac{v^2}{c^2}\right) = \frac{1}{2} \frac{v^2}{c^2} \\ &= \frac{GM}{2rc^2} \end{aligned}$$

This has a positive sign but represents a delay in the proper time of moving observers relative to static ones.

3.2.3. Velocity Correction Applied to Satellite Geolocation

To obtain the difference in time dilation between satellites and ground observers, we should use their velocity difference. However, due to the non-uniformity of surface velocities, we use only the orbital velocity relative to the ECI frame.

The ratio between satellite velocity v_{sat} and surface velocity v_{Earth} is:

(3.16)

$$\frac{v_{\text{sat}}}{v_{\text{Earth}}} = \frac{\omega_{\text{sat}}}{\omega_{\text{Earth}}} \frac{(r_t + h)}{r_t} = \frac{\omega_{\text{sat}}}{\omega_{\text{Earth}}} \left(1 + \frac{h}{r_t}\right)$$

where h is satellite altitude.

Since satellites have an [orbital period](#) of roughly half a day, $\omega_{\text{sat}} \approx 2 \times \omega_{\text{Earth}}$. Also, for GNSS, $h \gtrsim 3 \times r_t$. Thus:

(3.17)

$$\frac{v_{\text{sat}}}{v_{\text{Earth}}} \gtrsim 2 \times (1 + 3) = 8$$

Therefore:

(3.18)

$$\begin{aligned} \Delta\tau_{\text{velocity}} &= \frac{1}{2c^2} (v_{\text{sat}}^2 - v_{\text{Earth}}^2) \approx \\ &\frac{1}{2c^2} (v_{\text{sat}}^2 - \frac{v_{\text{sat}}^2}{64}) \end{aligned}$$

where the second term is only $\sim 1.5\%$ of the first and can be neglected:

(3.19)

$$\Delta\tau_{\text{velocity}} = \Delta\tau_{\text{vel sat}} - \Delta\tau_{\text{vel Earth}} \approx \Delta\tau_{\text{vel sat}} = \frac{1}{2c^2} v_{\text{sat}}^2 = \frac{GM}{2rc^2}$$

Using GRS80 values ([#4](#)):

(3.20)

$$\begin{aligned} v_{\text{Earth equator}} &= \omega_{\text{Earth}} \times r_{\text{equator}} \\ &= 7.292.115 \times 10^{-11} \\ &\times 6.378.137 = 465,1 \text{ m/s} \end{aligned}$$

For a satellite at 24,385 km altitude:

(3.21)

$$\begin{aligned} \varepsilon(\%) &= \frac{(v_{\text{sat}}^2 - v_{\text{Earth}}^2)}{v_{\text{sat}}^2} \times 100 = \\ &\frac{(3.600^2 - 465,1^2)}{3.600^2} \times 100 = 1,67\% \end{aligned}$$

Thus:

(3.22)

$$\Delta\tau_{\text{velocity}} = \frac{GM}{2(r_t + h)c^2}$$

or, using the Schwarzschild radius:

(3.23)

$$\Delta\tau_{\text{velocity}} = \frac{GM}{2(r_t + h)c^2} = \frac{r_s}{4(r_t + h)}$$

3.3. Relativistic Corrections Due to Gravity

We know that the proper time of two observers in a gravitational field flows differently depending on their difference in gravitational potential. We now want to compute these differences for Earth's gravitational field.

3.3.1. Relationship Between Proper Time and Gravitational Potential in Spherical Symmetry

We use the same working metric ([Schwarzschild](#)), expressed as:

(3.24)

$$\begin{aligned} ds^2 &= -c^2 \left(1 - \frac{r_s}{r}\right) dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + \\ &r^2 d\Omega^2 \quad \text{where } r_s = \frac{2GM}{c^2} \end{aligned}$$

Spherical static symmetry implies that the potential does not depend on angular

coordinates or on coordinate time—only on the radius. It is important to note that we are not trying to define the spacetime interval between two events at different radii, but rather the difference in proper-time flow relative to coordinate time at two different radii.

Since g_{tt} already relates proper time and coordinate time, setting $dr = d\Omega = 0$ yields:

(3.25)

$$d\tau^2 = \left(1 - \frac{r_s}{r}\right) dt^2 \Rightarrow \frac{d\tau}{dt} = \sqrt{\left(1 - \frac{r_s}{r}\right)}$$

Because r and t are independent variables, r is constant during integration over time. Therefore:

(3.26)

$$\Delta\tau_{\text{gravity}} = \sqrt{1 - \frac{r_s}{r}} \Delta t$$

For weak fields, we [approximate](#):

(3.27)

$$\Delta\tau_{\text{gravity}} = \sqrt{1 - \frac{r_s}{r}} \Delta t \approx \left(1 - \frac{r_s}{2r}\right) \Delta t$$

This expression allows us to compute the difference in proper-time flow between two radi r_a and r_b with $r_b > r_a$:

(3.28)

$$\begin{aligned} \Delta\tau_{\text{gravity}} &= \Delta\tau_{\text{grav } r_b} - \Delta\tau_{\text{grav } r_a} \\ &\approx \left(1 - \frac{r_s}{2r_b}\right) - \left(1 - \frac{r_s}{2r_a}\right) \\ &= \frac{r_s}{2r_a} - \frac{r_s}{2r_b} \end{aligned}$$

This can be generalized for any radius r and increment Δr :

(3.29)

$$\Delta\tau_{\text{gravity}} = \frac{r_s}{2r} - \frac{r_s}{2(r + \Delta r)}$$

3.3.2. Gravitational Correction Applied to Satellite Geolocation

We can rewrite expression 3.29 using the classical Newtonian potential $V = -\frac{GM}{r}$, substituting the Schwarzschild radius and assigning the values for Earth's radius r_t and the satellite orbital radius r_{sat} :

(3.30)

$$\begin{aligned} \Delta\tau_{\text{gravity}} &= \frac{r_s}{2r_t} - \frac{r_s}{2r_{\text{sat}}} = \frac{GM}{r_t c^2} - \frac{GM}{r_{\text{sat}} c^2} \\ &= \frac{V_t}{c^2} - \frac{GM}{r_{\text{sat}} c^2} \end{aligned}$$

Here,

V_t is the gravitational potential at Earth's surface. This substitution is convenient because both this value and Earth's mean radius are explicitly included in the GRS80 mode (#4).

3.3.3. Discriminating the Effects of Nearby Masses

It seems appropriate to check whether masses near the Earth influence the GNSS system by creating significant potential differences between the surface and the satellites beyond those created by the Earth's field. Fortunately, this is not the case, and we will verify that their influence is indeed negligible.

In the EFE in its general form:

(3.31)

$$R_{ab} - \frac{1}{2} R g_{ab} = G_{ab} = \frac{8\pi G}{c^4} T_{ab}$$

all coupled equations must be satisfied. For combined systems of n gravity sources, all equations of the following form would have to be satisfied:

(3.32)

$$G_{ab}^{(n)} = R_{ab}^{(n)} - \frac{1}{2} R^{(n)} g_{ab}^{(n)} = \frac{8\pi G}{c^4} T_{ab}^{(n)}$$

which, as we see, can never be generalized as a system of equations because the curvature tensors would belong to different metrics; thus, we cannot express the curvature as a linear expression of those generated by the different masses.

Verifying that we can discriminate the influence of other masses in the solar system thus becomes very difficult unless all fields are weak and we can sum potentials in a Newtonian manner. That is, by assuming the curved metric as the sum of a flat metric and a minimal perturbation that allows us to linearize it, in the form:

(3.33)

$$g_{ab} = \eta_{ab} + x_{ab} \text{ con } x_{ab} \ll \eta_{ab}$$

where η_{ab} is the flat Minkowski metric, g_{ab} represents the metric we want to linearize, and x_{ab} is considered the perturbation that adds curvature. In this way, the equations remain linearized in the weak-field regime.

The problem is simplified by confirming that all masses in the solar system generate weak fields and that their influence on Earth, which can now be summed, is negligible compared to the terrestrial field. Let us check these fields for each significant mass in the solar system.

The Sun's influence, by the equivalence principle, will only manifest in tidal forces. Given that these tidal forces manifest with a value inversely proportional to the cube of the distance and that the average of that distance is by definition 1 UA $\approx 150 \times 10^9$ m (#5, glossary), we can assume them to be negligible.

For the representative planets and satellites of the solar system, ordering their influence from greatest to least on the Earth's gravitational field—with the Earth's potential at its surface being $\Phi_{Earth} = -6,26 \times 10^7$ J/kg (#4, Derived Physical Constants) and according to JPL values (#5, Horizons, Astrodynamic Parameters)—we can construct the following table (Table 1)

Body	Min D (km)	Φ at D (J/kg)	Φ/Φ_{Earth}
Júpiter	$5,88 \times 10^8$	$-2,15 \times 10^5$	$3,44 \times 10^{-3}$
Saturno	$1,20 \times 10^9$	$-3,16 \times 10^4$	$5,06 \times 10^{-4}$
Luna	363.300	$-1,35 \times 10^4$	$2,16 \times 10^{-4}$
Venus	$3,82 \times 10^7$	$-8,55 \times 10^3$	$1,35 \times 10^{-4}$
Urano	$2,57 \times 10^9$	$-2,25 \times 10^3$	$3,60 \times 10^{-5}$
Neptuno	$4,30 \times 10^9$	$-1,59 \times 10^3$	$2,54 \times 10^{-5}$
Marte	$5,46 \times 10^7$	$-7,84 \times 10^2$	$1,25 \times 10^{-5}$
Mercurio	$7,73 \times 10^7$	$-2,85 \times 10^2$	$4,56 \times 10^{-6}$

Table 1

Where we can see that we can also neglect the potential differences they generate on the terrestrial field for the purposes of the GNSS system.

As a curiosity, although we can neglect their gravitational effects regarding the calculation of the temporal correction to be applied to the satellites, we must take

them into account in orbital calculations; in fact, in GNSS systems in use today, they are considered in the propagation of ephemerides.

3.4. Total relativistic correction

Since the proper time for any observer in the radial field leads due to its gravitational potential (the offset will have a positive sign) but lags due to its velocity (offset with a negative sign), the total necessary correction derived from relativistic effects for weak fields and circular orbits is:

(3.34)

$$\Delta\tau_{\text{total}} = \Delta\tau_{\text{gravity}} - \Delta\tau_{\text{velocity}} \approx \left(\frac{r_s}{2r} - \frac{r_s}{2(r + \Delta r)} \right) - \frac{r_s}{4(r + \Delta r)}$$

which can be expressed as:

(3.35)

$$\Delta\tau_{\text{total}} \approx \frac{r_s}{2r} - \frac{3r_s}{4(r + \Delta r)} = \frac{2GM}{rc^2} - \frac{3GM}{2(r + \Delta r)c^2}$$

Thus, particularizing the expression for the satellites relative to ground observers, we have:

(3.36)

$$\Delta\tau_{\text{total}} \approx \frac{V_t^2}{c^2} - \frac{3GM}{2r_{\text{sat}}c^2}$$

which is the expression that will provide the offset to be corrected in the satellites to avoid errors induced by relativistic effects.

3.5. Sensitivity analysis

We can perform a sensitivity analysis for all significant parameters to see how their variation influences the measurement of the passage of time in the case of two mobiles within a radial field, particularly in the time calculations required for geolocation systems.

3.5.1. Sensitivity analysis for velocity difference

Since the orbit of the satellites remains constant, we can assume that their orbital velocity is also constant. Similarly, we cannot expect changes in the Earth's rotational motion; therefore, the velocity difference between the satellites and the ground observers will remain relatively constant, in addition to all previous considerations regarding the ECI reference system and software correction.

Observers on the surface moving at high speeds may occur, but those corrections fall within the interferences to be analyzed in other software corrections that lie outside the scope of this document.

3.5.2. Sensitivity analysis for potential difference

Recovering the previous expressions after defining proper time τ such that $ds^2 = -c^2 d\tau^2$, and therefore:

(3.37)

$$d\tau^2 = \left(1 - \frac{r_s}{r}\right) dt^2 \Rightarrow \frac{d\tau}{dt} = \sqrt{\left(1 - \frac{r_s}{r}\right)}$$

we can recover the concept of [compactness](#) u and define the previous relationship as:

(3.38)

$$\frac{d\tau}{dt} = \sqrt{1 - u}$$

which provides the differential variation in proper time as a function of the compactness of the radial field, taking the form shown in next figure (figure 9)

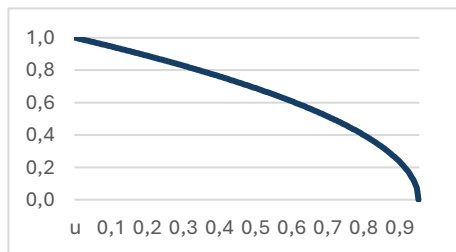


Figure 9

We can see that at points very far from the Schwarzschild radius, where compactness tends toward zero, time dilation

is not appreciable, while for objects near the Schwarzschild radius, time tends to dilate infinitely.

3.5.3. Application to the case of geolocation satellites

For the sensitivity analysis in satellite geolocation systems, we find ourselves in a case where the field is weak (compactness can be considered zero) and the relative velocity difference is constant; therefore, the following expression is completely valid:

(3.39)

$$\Delta\tau_x = \left(\frac{r_s}{2(r_t + x)} - \frac{r_s}{2((r_t + x) + h)} \right)$$

and it will provide, calling x the altitude above the Earth's surface, the variation in the measurement of proper time relative to the surface.

Assuming we move within the vicinity of the surface $x \ll r_t$ we have $r_t + x \approx r_t$; therefore, we can affirm that variation in altitude does not alter the measurement in positioning systems as long as we remain in the vicinity of the surface.

3.6. Verification of previous developments in operational systems

We want to validate the previous developments by introducing specific values for systems already in use on Earth. Thus, for GPS specifically, according to 3.36:

$$\Delta\tau_{\text{total}} \approx \frac{V_t}{c^2} - \frac{3GM}{2r_{\text{sat}} c^2}$$

With values

(#4) $V_t = 6\,263\,686,085 \times 10^{-8} \text{ m}^2/\text{s}^2$

(#6) $c = 299\,792\,458 \text{ m/s}$

(#4) $GM = 3\,986\,005 \times 10^8 \text{ m}^3/\text{s}^2$

(#6) $r_{\text{sat}} = 26\,559\,710 \text{ m}$ (real elliptical orbit semi-major axis)

the formula to apply is:

$$\Delta\tau_{\text{total}} \approx$$

$$\begin{aligned}
& \frac{6\,263\,686,0850 \times 10}{299\,792\,458^2} \\
& \quad - \frac{3 \times 3\,986\,005 \times 10^8}{2 \times 26\,559\,710 \times 299\,792\,458^2} \\
& = 6,9693 \times 10^{-10} - 2,5047 \times 10^{-10} \\
& = 4,4646 \times 10^{-10}
\end{aligned}$$

Consequently, the proper time of the satellites leads relative to the Earth's surface by $4,4646 \times 10^{-10} \text{ s}$ for every second that passes on the surface. That is, it will daily lead by:

$$\begin{aligned}
\Delta\tau_{\text{daily}} & \approx \\
(4,4646 \times 10^{-10} \times 3600 \times 24) \text{ s/day} \\
& = 38,574 \times 10^{-6} \text{ s/día} = 38.5 \text{ } \mu\text{s/day}
\end{aligned}$$

This is exactly the correction applied in real GPS according to its ICD (#6, 3.3.1.1 Frequency Plan) , which states that if the nominal frequency of GPS clocks on the surface is 10.23 MHz, the frequency on the satellites must be adjusted to:

$$\begin{aligned}
f' & = (1 - \Delta\tau_{\text{total}}) \times f \\
& = (1 - 4,4647 \times 10^{-10}) \times 10,23 \text{ MHz} \\
& = 10,2299999954326 \text{ MHz}
\end{aligned}$$

This adjustment is introduced into GPS clocks before launch. It has been verified that the expression also returns correct values for other systems, although in Galileo the time correction is performed via software and not by modifying the onboard clock offset. The results are shown in the following table (Table 2):

GPS	
r_{sat}	26 559 710 (#6)
$\Delta\tau_{\text{total}}$ calculated	$4,4646 \times 10^{-10}$
$\Delta\tau_{\text{total}}$ sources	$4,4647 \times 10^{-10}$ (#6)
$\Delta\tau_{\text{daily}}$	38,5 $\mu\text{s/día}$
GLONASS	
r_{sat}	25 478 136 (#7)
$\Delta\tau_{\text{total}}$ calculated	$4,3581 \times 10^{-10}$
$\Delta\tau_{\text{total}}$ sources	$4,36 \times 10^{-10}$ (#7)
$\Delta\tau_{\text{daily}}$	37,6 $\mu\text{s/día}$
Galileo	
r_{sat}	29 601 297 (#8)
$\Delta\tau_{\text{total}}$ calculated	$4,7216 \times 10^{-10}$
$\Delta\tau_{\text{total}}$ sources	$4,7219 \times 10^{-10}$ (#3)
$\Delta\tau_{\text{daily}}$	40,7 $\mu\text{s/día}$

Table 2

As a curiosity and proof of the previous conclusions regarding the sensitivity of the GPS system, the error introduced if we wanted to use it at an altitude of 10 km (the

flight altitude of commercial aviation) would be:

$$\begin{aligned}
\Delta\tau_x & = \left(\frac{r_s}{2(r_t + x)} - \frac{r_s}{2((r_t + x) + h)} \right) \\
& = \frac{8,8701 \times 10^{-3}}{2} \left(\frac{1}{(6\,371 + 10) \times 10^3} \right. \\
& \quad \left. - \frac{1}{(6\,371 + 10 + 20\,200) \times 10^3} \right) \\
& = 5,2818 \times 10^{-10}
\end{aligned}$$

Which, as we see, is equivalent to the calculation on the surface by $\frac{5,2818 \times 10^{-10}}{5,2922 \times 10^{-10}} = 99,80\%$, i.e., to the relativity correction already predetermined in the GPS system. Also, as a curiosity, if proper time were not corrected in the satellites, the accumulated daily error in each distance measured relative to GPS satellites would be:

$$\begin{aligned}
\Delta d_{\text{daily}} & \approx 38,5145 \times 10^{-6} \text{ s/day} \times c \\
& = 11\,546.35 \text{ m/day} = 11.5 \text{ km each día}
\end{aligned}$$

4. Anexos y demostraciones

Vamos a introducir algunas notas y demostraciones generales de mecánica analítica y relatividad general que pueden ser útiles en el contexto de este trabajo así como otras directamente relacionadas con la navegación por satélite.

4.1. Orbits in GNSS Satellites

GNSS satellites follow free-fall orbits with typically low eccentricity ($\lesssim 0,02$) allowing us to treat them as perfect circles for the purpose of computing linear velocity, i.e., the tangential component of the trajectory. This is not the case for their ephemerides, where position calculations would be incorrect if the exact eccentricity were not implemented..

Since one of the fundamental design parameters is to cover as much of the Earth's surface as possible with precision while minimizing other costs—such as the number of satellites required—and since frequent updates from the ground are necessary, medium Earth orbits are defined below approximately 35,000 km altitude, around 20,000 km. This avoids the inner Van Allen belt, which would destroy electronics and atomic clocks and is the main reason GNSS systems can exist.

We know that the linear velocity is:

We know that the [tangential velocity](#) is $v = \sqrt{\frac{GM}{r}}$ and can also be expressed as $v = \omega \times r$. We also know that ω is the angular distance traveled per unit time, so for a time T in which a full circular trajectory is completed:

(3.40)

$$v = \omega r = \frac{2\pi}{T} r$$

Equating both expressions:

(3.41)

$$v = \sqrt{\frac{GM}{r}} = \omega \times r = \frac{2\pi}{T} r \Rightarrow r = \sqrt[3]{\frac{GMT^2}{4\pi^2}}$$

This gives the relationship between the radius of a stable circular orbit and its period. For current global GNSS systems, we obtain (Table 3):

GNSS	Period (T)	r_{sat} (km)	h (km)
GPS (#6)	11h 58min	26.560	20.182
GLONASS (#7)	11h 15min	25.510	19.140
Galileo (#8)	14h 05min	29.593	23.222
BeiDou (#9)	12h 53min	27.906	21.535

Table 3

These orbits also satisfy other design constraints beyond the absence of drag required for stable circular orbits with constant angular momentum. Using higher radii would require higher signal power, and the cost of placing satellites in such orbits would also be greater.

4.2. Treatment of the Relativistic Correction in Technical Publications (Virial)

We observe that the work carried out in this document is a sufficiently accurate approximation to the published values for GNSS systems in use.

A notable point is the formula used in Ashby (#2, equation 35), which is identical to the one used here, substituting the radius with the semi-major axis of the ellipse and using properties of the WGS84 geoid:

$$\Delta\tau = \frac{3GM_E}{2ac^2} + \frac{\Phi_0}{c^2}$$

This corresponds to the formulation developed in (3.36), where we changed signs and assigned opposite signs to the velocity and gravitational corrections:

$$\Delta\tau_{\text{total}} \approx \frac{V_t}{c^2} - \frac{3GM}{2r_{\text{sat}} c^2}$$

It is particularly notable that the correction is the same even though Ashby uses conservation of energy in the Keplerian elliptical trajectory, while here we assumed a perfect circular orbit. Ashby uses the real trajectory and works with eccentricity (denoted e , defined as the ratio between the focal distance and the semi-major axis), even if it is minimal (GPS max eccentricity 0.02; GLONASS, Galileo, BeiDou: 0.000–0.001, #6, #7, #8 y #9).

This coincidence arises from the virial theorem, which states that a time-averaged Keplerian elliptical orbit has the same mean kinetic and potential energy as a circular orbit whose radius equals the ellipse's semi-major axis.

The analytical mechanics required to incorporate eccentricity has been omitted, but any interested reader is encouraged to work through it, as well as the Sagnac correction (much simpler), to better understand relativistic effects in GNSS systems, both at the satellite and ground-receiver level.

4.3. Algunas aproximaciones de Taylor

En trabajos de relatividad especial y general vamos a encontrarnos con frecuencia con dos formas para las que nos será útil emplear las siguientes aproximaciones de Taylor

$$f(x) = \sqrt{1-x} \approx 1 - \frac{1}{2}x \quad \text{válida para } |x| \ll 1$$

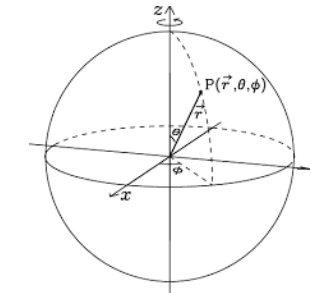
$$f(x) = (1+x)^n \approx 1 + nx \quad \text{válida para } |x| \ll 1$$

En relatividad general x aparece como la [compacidad](#) $u = \frac{r_s}{r}$ y puede ser aproximada a cero para campos débiles (por tener un orden de magnitud $u \sim 10^{-27-n+b}$ donde n es el orden de magnitud del radio y b el de la masa) con lo que la expresión es perfectamente válida.

En relatividad especial x aparece como la relación $\left(\frac{v}{c}\right)^2$ restando en la raíz del denominador de la transformación de Lorentz que hace la expresión perfectamente válida para velocidades mucho menores que c .

4.4. Solución de Schwarzschild para simetrías esféricas estáticas

Vamos a demostrar la solución de Schwarzschild partiendo de la métrica de Minkowski en coordenadas esféricas de la forma



$r \in [0, \infty)$
 $\theta \in [0, \pi]$
 $\phi \in [0, 2\pi)$

$$ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2$$

$$\begin{cases} x = r \sin \theta \cos \phi \\ y = r \sin \theta \sin \phi \\ z = r \cos \theta \end{cases}$$

$$ds^2 = -c^2 dt^2 + dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2$$

que veremos escrita con frecuencia de la siguiente manera para resumir las coordenadas angulares

$$ds^2 = -c^2 dt^2 + dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\phi^2) = -c^2 dt^2 + dr^2 + r^2 d\Omega^2$$

Y formularemos una simetría esférica similar donde todas las variables libres se expresen como funciones $f_i(r)$ del radio y sólo del radio (la expresión para la coordenada angular permanece inalterada para mantener la simetría)

$$ds^2 = -c^2 f_t(r) dt^2 + f_r(r) dr^2 + r^2 d\Omega^2$$

Se mostrará conveniente escribir las funciones $f_i(r)$ de la expresión anterior como funciones exponenciales multiplicadas por 2 para simplificar todos los cálculos posteriores y por tanto

$$ds^2 = -c^2 e^{2f_t(r)} dt^2 + e^{2f_r(r)} dr^2 + r^2 d\Omega^2$$

Queremos calcular las funciones $f_i(r)$, es decir, los términos $e^{f_i(r)}$ que hagan que la métrica esférica satisfaga la ecuación de campo de Einstein en el vacío ($T_{ab} = 0$) y sin cte cosmológica Λ (porque estamos calculando la solución simple con un espacio-tiempo asintóticamente plano)

$$R_{ab} - \frac{1}{2} R g_{ab} = \frac{8\pi G}{c^4} T_{ab} = 0$$

donde vemos que $R_{ab} = 0$ es condición necesaria para satisfacer la ecuación (si $R_{ab} = 0$ su escalar R será también nulo para cualquier métrica g_{ab}). Por tanto el problema se limita a encontrar el tensor R_{ab} para la métrica postulada y conocer el valor que tienen que satisfacer las funciones $f_i(r)$ para que $R_{ab} = 0$

4.4.1. Cálculo de símbolos de Christoffel

Necesitaremos hallar todos los símbolos de Christoffel de esa métrica necesarios para calcular las componentes del tensor de Ricci a igualar a cero y lo haremos a través del principio de mínima acción.

Para ello definiremos un lagrangiano de la forma $L = \left(\frac{ds}{d\lambda}\right)^2$ con un parámetro afín λ en el que L se mantiene cte $\frac{\partial L}{\partial \lambda} = 0$. Expresando la métrica en notación tensorial como $ds^2 = g_{ab} dx^a dx^b$ y usando la notación $\dot{x} = \frac{dx}{d\lambda}$ se puede escribir el lagrangiano de la forma $L = \left(\frac{ds}{d\lambda}\right)^2 = g_{ab} \dot{x}^a \dot{x}^b$

Ese lagrangiano aplicado a la métrica en forma exponencial que habíamos propuesto resulta

$$L = -c^2 e^{2f_t(r)} \dot{t}^2 + e^{2f_r(r)} \dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\phi}^2$$

Donde hemos vuelto a desarrollar la componente angular para poder hallar los símbolos de Christoffel asociados a θ y ϕ . Desarrollaremos todos los términos de la expresión de mínima-acción de Euler-Lagrange con ese lagrangiano

$$\frac{d}{d\lambda} \left(\frac{\partial L}{\partial \dot{x}^a} \right) - \frac{\partial L}{\partial x^a} = 0$$

y compararemos los términos resultantes con la ecuación geodésica en

forma canónica para obtener los Γ^a de cada coordenada

$$\ddot{x}^a + \Gamma_{bc}^a \dot{x}^b \dot{x}^c = 0$$

Desarrollando esos Γ_{bc}^a para nuestro espacio de cuatro dimensiones para cada coordenada x^a según las expresiones de todos los otros pares posibles de coordenadas x^a y x^b de la métrica la comparación de términos será inmediata.

Desarrollo para la coordenada t

$$L = -c^2 e^{2f_t(r)} \dot{t}^2 + e^{2f_r(r)} \dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\phi}^2$$

Recordemos que todas las variables son independientes y sus derivadas también (espacio de estados)

$$\begin{aligned} \frac{d}{d\lambda} \left(\frac{\partial L}{\partial \dot{t}} \right) &= \frac{d}{d\lambda} (-2c^2 e^{2f_t(r)} \dot{t}) \\ &= \frac{d}{d\lambda} (-2c^2 e^{2f_t(r)}) \dot{t} - 2c^2 e^{2f_t(r)} \ddot{t} \end{aligned}$$

Aplicando la regla de la cadena $\frac{d}{d\lambda} [f(r(\lambda))] = \frac{df}{dr} \cdot \frac{dr}{d\lambda}$ que podemos escribir como $\frac{df}{dr} \dot{r} = f' \dot{r}$ pero no como $\dot{f} \dot{r}$ porque f no se deriva con respecto al parámetro afín tenemos que

$$\begin{aligned} \frac{d}{d\lambda} \left(\frac{\partial L}{\partial \dot{t}} \right) &= [-4c^2 f'_t(r) e^{2f_t(r)} \dot{r}] \dot{t} - 2c^2 e^{2f_t(r)} \ddot{t} \\ \frac{\partial L}{\partial t} &= 0 \end{aligned}$$

Por tanto la ecuación geodésica se puede escribir como

$$[-4f'_t(r) c^2 e^{2f_t(r)} \dot{r}] \dot{t} - 2c^2 e^{2f_t(r)} \ddot{t} = 0$$

Dividiendo entre $-2c^2 e^{2f_t(r)}$ tenemos que

$$\ddot{t} + 2f'_t(r) \dot{r} \dot{t} = 0 \text{ que comparada con } \ddot{x}^a + \Gamma_{bc}^a \dot{x}^b \dot{x}^c = 0$$

Nos indica que $\Gamma_{rt}^t = \Gamma_{tr}^t = f'_t(r)$ y el resto de símbolos de Christoffel en t son nulos.

Conservación de la energía

Aquí es importante notar además que $\frac{\partial L}{\partial t} = 0$ con lo que el lagrangiano se conserva en t para cualquier parámetro afín y que de

dicha conservación resulta además que $\frac{d}{d\lambda} \left(\frac{\partial L}{\partial \dot{t}} \right) = 0$ donde $\frac{\partial L}{\partial \dot{t}} = -2c^2 e^{2f_t(r)} \dot{t}$

Por tanto $-2c^2 e^{2f_t(r)} \dot{t}$ es una cte con respecto a cualquier parámetro afín. A esta magnitud cte la llamaremos energía por unidad de masa ϵ o energía específica y a su constancia conservación de la energía. Y para su uso general eliminaremos todas las ctes cuya derivada es cero para no dimensionar mal las ecuaciones en que la utilizemos de la forma

$$\epsilon = e^{2f_t(r)} \dot{t} \Leftrightarrow \frac{\partial L}{\partial t} = 0 \Leftrightarrow \frac{\partial L}{\partial t} = \epsilon$$

Dicha conservación se mantiene ante [cambios de coordenadas](#) en ϕ y t aunque no en cambios de coordenadas en r o θ .

Desarrollo para la coordenada r

$$L = -c^2 e^{2f_t(r)} \dot{t}^2 + e^{2f_r(r)} \dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\phi}^2$$

Recordemos que todas las variables son independientes y sus derivadas también (espacio de estados)

$$\frac{d}{d\lambda} \left(\frac{\partial L}{\partial \dot{r}} \right) = \frac{d}{d\lambda} (2e^{2f_r(r)} \dot{r}) = \frac{d}{d\lambda} (2e^{2f_r(r)}) \dot{r} + 2e^{2f_r(r)} \ddot{r} =$$

$$\frac{d}{d\lambda} \left(\frac{\partial L}{\partial \dot{r}} \right) = 4f'_r(r) e^{2f_r(r)} \dot{r}^2 + 2e^{2f_r(r)} \ddot{r}$$

$$\begin{aligned} \frac{\partial L}{\partial r} &= 2r(\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2) \\ &\quad - 2c^2 f'_t(r) e^{2f_t(r)} \dot{t}^2 \\ &\quad + 2f'_r(r) e^{2f_r(r)} \dot{r}^2 \end{aligned}$$

Por tanto la ecuación geodésica se puede escribir y desarrollar como

$$\begin{aligned} 4f'_r(r) e^{2f_r(r)} \dot{r}^2 + 2e^{2f_r(r)} \ddot{r} \\ - [2r(\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2) - 2c^2 f'_t(r) e^{2f_t(r)} \dot{t}^2 + 2f'_r(r) e^{2f_r(r)} \dot{r}^2] = 0 \end{aligned}$$

$$\begin{aligned} 4f'_r(r) e^{2f_r(r)} \dot{r}^2 + 2e^{2f_r(r)} \ddot{r} - 2r(\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2) + 2c^2 f'_t(r) e^{2f_t(r)} \dot{t}^2 - 2f'_r(r) e^{2f_r(r)} \dot{r}^2 = 0 \end{aligned}$$

$$\begin{aligned} 2e^{2f_r(r)} \ddot{r} + 4f'_r(r) e^{2f_r(r)} \dot{r}^2 - 2r(\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2) + 2c^2 f'_t(r) e^{2f_t(r)} \dot{t}^2 - 2f'_r(r) e^{2f_r(r)} \dot{r}^2 = 0 \end{aligned}$$

$$\begin{aligned} 2e^{2f_r(r)} \ddot{r} + 2f'_r(r) e^{2f_r(r)} \dot{r}^2 - 2r(\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2) + 2c^2 f'_t(r) e^{2f_t(r)} \dot{t}^2 = 0 \end{aligned}$$

Dividiendo entre $2e^{2f_r(r)}$ tenemos que

$$\ddot{r} + f'_r(r) \dot{r}^2 - r(e^{-2f_r(r)})(\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2) + c^2 f'_t(r) e^{2f_t(r)-2f_r(r)} \dot{t}^2 = 0$$

Que comparada con $\ddot{x}^a + \Gamma_{bc}^a \dot{x}^b \dot{x}^c = 0$ nos indica que

$$\Gamma_{rr}^r = f'_r(r)$$

$$\Gamma_{\theta\theta}^r = -r e^{-2f_r(r)}$$

$$\Gamma_{\phi\phi}^r = -r \sin^2 \theta e^{-2f_r(r)}$$

$$\Gamma_{tt}^r = c^2 f'_t(r) e^{2f_t(r)-2f_r(r)}$$

El resto de símbolos de Christoffel en r son nulos.

Desarrollo para la coordenada θ

$$L = -c^2 e^{2f_t(r)} \dot{t}^2 + e^{2f_r(r)} \dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\phi}^2$$

Recordemos que todas las variables son independientes y sus derivadas también (espacio de estados)

$$\frac{d}{d\lambda} \left(\frac{\partial L}{\partial \dot{\theta}} \right) = \frac{d}{d\lambda} (2r^2 \dot{\theta}) = \frac{d}{d\lambda} (2r^2) \dot{\theta} + 2r^2 \ddot{\theta} = 4r \dot{r} \dot{\theta} + 2r^2 \ddot{\theta}$$

$$\frac{\partial L}{\partial \theta} = 2r^2 \sin \theta \cos \theta \dot{\phi}^2 = r^2 \sin 2\theta \dot{\phi}^2$$

Por tanto la ecuación geodésica se puede escribir como

$$4r \dot{r} \dot{\theta} + 2r^2 \ddot{\theta} - r^2 \sin 2\theta \dot{\phi}^2 = 0$$

Dividiendo entre $2r^2$ tenemos que

$$\ddot{\theta} + (2/r) \dot{r} \dot{\theta} - (1/2) \sin 2\theta \dot{\phi}^2 = 0$$

Que comparada con $\ddot{x}^a + \Gamma_{bc}^a \dot{x}^b \dot{x}^c = 0$ nos indica que

$$\Gamma_{r\theta}^{\theta} = \Gamma_{\theta r}^{\theta} = \frac{1}{r}$$

$$\Gamma_{\phi\phi}^{\theta} = -(1/2) \sin 2\theta$$

El resto de símbolos de Christoffel en θ son nulos.

Desarrollo para la coordenada ϕ

$$L = -c^2 e^{2f_t(r)} \dot{t}^2 + e^{2f_r(r)} \dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\phi}^2$$

Recordemos que todas las variables son independientes y sus derivadas también (espacio de estados)

$$\begin{aligned} \frac{d}{d\lambda} \left(\frac{\partial L}{\partial \dot{\phi}} \right) &= \frac{d}{d\lambda} (2r^2 \sin^2 \theta \dot{\phi}) \\ &= \frac{d}{d\lambda} (2r^2 \sin^2 \theta) \dot{\phi} \\ &\quad + 2r^2 \sin^2 \theta \ddot{\phi} \end{aligned}$$

$$\begin{aligned} &= (4r^2 \sin \theta \cos \theta \dot{\theta} + 4r \dot{r} \sin^2 \theta) \dot{\phi} + \\ &2r^2 \sin^2 \theta \ddot{\phi} = 4r^2 \sin \theta \cos \theta \dot{\theta} \dot{\phi} + \\ &4r \dot{r} \sin^2 \theta \dot{\phi} + 2r^2 \sin^2 \theta \ddot{\phi} \end{aligned}$$

$$\frac{\partial L}{\partial \phi} = 0$$

Por tanto la ecuación geodésica se puede escribir como

$$4r^2 \sin \theta \cos \theta \dot{\theta} \dot{\phi} + 4r \dot{r} \sin^2 \theta \dot{\phi} + 2r^2 \sin^2 \theta \ddot{\phi} = 0$$

Dividiendo entre $2r^2 \sin^2 \theta$ tenemos que

$$\ddot{\phi} + 2 \cot \theta \dot{\theta} \dot{\phi} + 2(1/r) \dot{r} \dot{\phi} = 0$$

Que comparada con $\ddot{x}^a + \Gamma_{bc}^a \dot{x}^b \dot{x}^c = 0$ nos indica que

$$\Gamma_{\theta\phi}^{\phi} = \Gamma_{\phi\theta}^{\phi} = \cot \theta$$

$$\Gamma_{r\phi}^{\phi} = \Gamma_{\phi r}^{\phi} = \frac{1}{r}$$

El resto de símbolos de Christoffel en ϕ son nulos.

Conservación del momento angular

Aquí es importante notar además que $\frac{\partial L}{\partial \phi} = 0$ con lo que el lagrangiano se conserva en ϕ para cualquier parámetro afín y que de dicha conservación resulta además que $\frac{d}{d\lambda} \left(\frac{\partial L}{\partial \dot{\phi}} \right) = 0$ donde $\frac{\partial L}{\partial \dot{\phi}} = 2r^2 \sin^2 \theta \dot{\phi}$

Por tanto $2r^2 \sin^2 \theta \dot{\phi}$ es una cte con respecto a cualquier parámetro afín. A esta magnitud cte la llamaremos momento angular por unidad de masa o momento específico j (para distinguirla del lagrangiano L) y a su constancia conservación del momento angular. Y para su uso general

eliminaremos todas las ctes cuya derivada es cero para no dimensionar mal las ecuaciones en que la utilizemos de la forma

$$j = r^2 \sin^2 \theta \dot{\phi} \Leftrightarrow \frac{\partial L}{\partial \dot{\phi}} = 0 \Leftrightarrow \frac{\partial L}{\partial \phi} = j$$

Dicha conservación se mantiene ante [cambios de coordenadas](#) en ϕ y t aunque no en cambios de coordenadas en r o θ .

4.4.2. Cálculo de las componentes del tensor de Ricci R_{ab}

Ya estamos en posición de obtener las componentes de Ricci e igualarlas a cero para obtener las funciones exponenciales de la métrica propuesta con la siguiente tabla resumen de símbolos de Christoffel no nulos

Símbolos de Christoffel en Schwarzschild		
$\Gamma_{rt}^t = \Gamma_{tr}^r = f'_t(r)$	$\Gamma_{\phi\phi}^r = -r \sin^2 \theta e^{-2f_r(r)}$	$\Gamma_{tt}^r = c^2 f'_t(r) e^{2f_t(r)-2f_r(r)}$
$\Gamma_{rr}^r = f'_r(r)$	$\Gamma_{\theta\theta}^r = -r e^{-2f_r(r)}$	$\Gamma_{\theta\phi}^\phi = \Gamma_{\phi\theta}^\phi = \cot \theta$
$\Gamma_{r\theta}^\theta = \Gamma_{\theta r}^\theta = \frac{1}{r}$	$\Gamma_{\phi\phi}^\theta = -(1/2) \sin 2\theta$	$\Gamma_{r\phi}^\phi = \Gamma_{\phi r}^\phi = \frac{1}{r}$

Vamos a desarrollar sólo una de las componentes del tensor para mostrar la complejidad del cálculo y por este motivo este será el único componente explícito de R_{ab} calculado en este documento. Para la solución completa de Schwarzschild introduciremos directamente los valores de R_{ab} que se pueden obtener de múltiples fuentes por ser este cálculo frecuente en problemas de relatividad general para la métrica propuesta.

Desarrollo particular para R_{tt}

$$R_{tt} = \partial_r \Gamma_{tt}^r + \Gamma_{tt}^r (\Gamma_{rr}^r + \Gamma_{r\theta}^\theta + \Gamma_{r\phi}^\phi) - \Gamma_{tr}^t \Gamma_{tt}^r$$

Tenemos que la única derivada parcial anterior corresponde a

$$\begin{aligned} \partial_r \Gamma_{tt}^r &= \partial_r (c^2 f'_t(r) e^{2f_t(r)-2f_r(r)}) \\ &= c^2 [f''_t(r) + 2(f'_t(r))^2 \\ &\quad - 2f'_t(r)f'_r(r)] e^{2f_t(r)-2f_r(r)} \end{aligned}$$

Sustituyendo y desarrollando tenemos que

$$\begin{aligned} R_{tt} &= c^2 [f''_t(r) + 2(f'_t(r))^2 \\ &\quad - 2f'_t(r)f'_r(r)] e^{2f_t(r)-2f_r(r)} \\ &\quad + c^2 f'_t(r) e^{2f_t(r)-2f_r(r)} \left[(f'_r(r)) + \frac{1}{r} + \frac{1}{r} \right] \\ &\quad - c^2 f'_t(r)^2 e^{2f_t(r)-2f_r(r)} \\ R_{tt} &= c^2 e^{2f_t(r)-2f_r(r)} \left[f''_t(r) + 2(f'_t(r))^2 \right. \\ &\quad - 2f'_t(r)f'_r(r) \\ &\quad + f'_t(r) \left((f'_r(r)) + \frac{1}{r} + \frac{1}{r} \right) \\ &\quad \left. - f'_t(r)^2 \right] \end{aligned}$$

$$R_{tt} = c^2 e^{2f_t(r)-2f_r(r)} \left[f''_t(r) + (f'_t(r))^2 - 2f'_t(r)f'_r(r) + f'_t(r) \left(\frac{rf'_r(r) + 2}{r} \right) \right]$$

$$R_{tt} = c^2 e^{2f_t(r)-2f_r(r)} \left[f''_t(r) + f'_t(r)^2 - 2f'_t(r)f'_r(r) + f'_t(r)f'_r(r) + \frac{2}{r} f'_t(r) \right]$$

$$R_{tt} = c^2 e^{2f_t(r)-2f_r(r)} \left[f''_t(r) + f'_t(r)^2 - f'_t(r)f'_r(r) + \frac{2}{r} f'_t(r) \right]$$

Que añadida al resto de componentes no nulas para el Ricci que podemos extraer de diferentes fuentes ([#1](#), ecuaciones 5.14) tenemos

$$R_{tt} = c^2 e^{2f_t(r)-2f_r(r)} \left[f''_t(r) + f'_t(r)^2 - f'_t(r)f'_r(r) + \frac{2}{r} f'_t(r) \right]$$

$$R_{rr} = f''_t(r) + (f'_t(r))^2 - f'_t(r)f'_r(r) - \frac{2f'_r(r)}{r}$$

$$R_{\theta\theta} = e^{-2f_r(r)} [1 + r(f'_t(r) - f'_r(r))] - 1$$

$$R_{\phi\phi} = \sin^2 \theta \cdot R_{\theta\theta} = \sin^2 \theta (e^{-2f_r(r)} [1 + r(f'_t(r) - f'_r(r))] - 1)$$

Donde se observa que todas las componentes cruzadas desaparecen y que las que no son nulas dependen del radio y sólo del radio, lo que es consistente con el supuesto inicial de simetría esférica con una

métrica diagonal dependiente del radio y sólo del radio.

4.4.3. Integración de R_{ab} en las ECE

4.4.3.1. Componentes individuales de R_{ab} en las ECE

Ahora igualaremos todas las componentes no nulas de $R_{ab} = 0$ para obtener las funciones $f_r(r)$ y $f_t(r)$ buscadas

$$R_{tt} = 0 \Rightarrow c^2 e^{2f_t(r)-2f_r(r)} \left[f_t''(r) + f_t'(r)^2 - f_t'(r)f_r'(r) + \frac{2}{r}f_t'(r) \right] = 0$$

$$R_{rr} = 0 \Rightarrow f_t''(r) + (f_t'(r))^2 - f_t'(r)f_r'(r) - \frac{2f_r'(r)}{r} = 0$$

$$R_{\theta\theta} = 0 \Rightarrow e^{-2f_r(r)} [1 + r(f_t'(r) - f_r'(r))] - 1 = 0$$

$$R_{\phi\phi} = 0 \Rightarrow \sin^2 \theta \cdot R_{\theta\theta} = \sin^2 \theta (e^{-2f_r(r)} [1 + r(f_t'(r) - f_r'(r))] - 1) = 0$$

De todas las ecuaciones acopladas anteriores escogeremos primero $R_{\theta\theta} = 0$ porque es la que implica consecuencias inmediatas para resolver el sistema.

$$R_{\theta\theta} = 0 \Rightarrow e^{-2f_r(r)} [1 + r(f_t'(r) - f_r'(r))] = 1$$

$$\Rightarrow [1 + r(f_t'(r) - f_r'(r))] = e^{2f_r(r)}$$

$$\Rightarrow r(f_t'(r) - f_r'(r)) = e^{2f_r(r)} - 1$$

$$(R_{\theta\theta} = 0) \Rightarrow f_t'(r) - f_r'(r) = \frac{e^{2f_r(r)} - 1}{r}$$

4.4.3.2. Comparación de componentes R_{tt} y R_{rr}

Mostrando todas las componentes en una única expresión tenemos

$$R_{tt} + R_{rr} + R_{\theta\theta} + R_{\phi\phi} = 0$$

o normalizando en el escalar

$$\begin{aligned} \Rightarrow R = g^{ab} R_{ab} &= g^{tt} R_{tt} + g^{rr} R_{rr} + g^{\theta\theta} R_{\theta\theta} \\ &+ g^{\phi\phi} R_{\phi\phi} \\ &= \frac{R_{tt}}{g_{tt}} + \frac{R_{rr}}{g_{rr}} + \frac{R_{\theta\theta}}{g_{\theta\theta}} + \frac{R_{\phi\phi}}{g_{\phi\phi}} \\ &= 0 \end{aligned}$$

Al haber planteado una simetría esférica donde $g_{\theta\theta}$ y $g_{\phi\phi}$ no contienen las

funciones $f_t(r)$ y $f_r(r)$ que queremos deducir no podremos utilizar sus componentes, pero sí las de t y r , de la forma

$$\frac{R_{tt}}{g_{tt}} + \frac{R_{rr}}{g_{rr}} = g^{tt} R_{tt} + g^{rr} R_{rr} = 0$$

y sustituyendo los valores de las expresiones individuales en esta expresión

$$\begin{aligned} &\frac{c^2 e^{2f_t(r)-2f_r(r)} \left[f_t''(r) + f_t'(r)^2 - f_t'(r)f_r'(r) + \frac{2}{r}f_t'(r) \right]}{-c^2 e^{2f_t(r)}} \\ &+ \frac{R_{rr} f_t''(r) + (f_t'(r))^2 - f_t'(r)f_r'(r) - \frac{2f_r'(r)}{r}}{e^{2f_r(r)}} \\ &= 0 \\ &\frac{-e^{2f_t(r)-2f_r(r)} \left[f_t''(r) + f_t'(r)^2 - f_t'(r)f_r'(r) + \frac{2}{r}f_t'(r) \right]}{e^{2f_t(r)}} \\ &+ \frac{f_t''(r) + (f_t'(r))^2 - f_t'(r)f_r'(r) - \frac{2f_r'(r)}{r}}{e^{2f_r(r)}} \\ &= 0 \\ &-e^{-2f_r(r)} \left[f_t''(r) + f_t'(r)^2 - f_t'(r)f_r'(r) + \frac{2}{r}f_t'(r) \right] \\ &- e^{2f_r(r)} \left[f_t''(r) + (f_t'(r))^2 - f_t'(r)f_r'(r) - \frac{2f_r'(r)}{r} \right] = 0 \end{aligned}$$

y cancelando término a término en $e^{-2f_r(r)}$

$$-e^{-2f_r(r)} \left[+ \frac{2f_t'(r)}{r} \right] - e^{2f_r(r)} \left[- \frac{2f_r'(r)}{r} \right] = 0$$

y como la exponencial nunca puede ser cero tenemos que

$$\frac{2}{r} [f_t'(r) + f_r'(r)] = 0$$

y por tanto

$$f_t'(r) + f_r'(r) = 0$$

$$\left(\frac{R_{tt}}{g_{tt}} + \frac{R_{rr}}{g_{rr}} = 0 \right) \Rightarrow f_t'(r) = -f_r'(r)$$

4.4.3.3. Integración de los desarrollos anteriores

Así tenemos que aplicando las dos expresiones que hemos concluido anteriormente tenemos que

$$(R_{\theta\theta} = 0) \Rightarrow f'_t(r) - f'_r(r) = \frac{e^{2f_r(r)} - 1}{r} \quad \text{y}$$

$$\left(\frac{R_{tt}}{g_{tt}} + \frac{R_{rr}}{g_{rr}} = 0\right) \Rightarrow f'_t(r) = -f'_r(r)$$

$$\Rightarrow f'_t(r) - f'_r(r) = \frac{e^{2f_r(r)} - 1}{r}$$

$$\Rightarrow 2rf'_t(r) = e^{-2f_t(r)} - 1$$

que es una ecuación ordinaria de primer orden con solución

$$f_t(r) = \frac{1}{2} \ln \left(1 - \frac{C}{r}\right) \text{ donde } C \text{ es una cte de integración}$$

y según nuestra suposición $f'_t(r) = -f'_r(r)$ tenemos que

$$f_r(r) = -\frac{1}{2} \ln \left(1 - \frac{C}{r}\right)$$

Es conveniente notar ya que las expresiones anteriores se pueden escribir exponencialmente como

$$e^{2f_t(r)} = 1 - \frac{C}{r} \text{ y } e^{2f_r(r)} = \frac{1}{1 - \frac{C}{r}}$$

y por tanto la métrica inicial puede ser escrita como

$$ds^2 = -c^2 e^{2f_t(r)} dt^2 + e^{2f_r(r)} dr^2 + r^2 d\Omega^2$$

$$ds^2 = -c^2 \left(1 - \frac{C}{r}\right) dt^2 + \left(\frac{1}{1 - \frac{C}{r}}\right) dr^2 + r^2 d\Omega^2$$

$$ds^2 = -c^2 \left(1 - \frac{C}{r}\right) dt^2 + \left(1 - \frac{C}{r}\right)^{-1} dr^2 + r^2 d\Omega^2$$

Donde nos queda por determinar el valor de la cte de integración C para las condiciones del problema.

4.4.4. Determinación de la masa (límite newtoniano)

Para ello vamos a hacer tres suposiciones basadas en la mecánica clásica.

La primera es suponer que esa cte de integración C es directamente proporcional a la masa que genera el campo, de forma que $C \propto M$

La segunda es la asunción de que cuando r tiende a infinito la métrica toma la

forma de un espacio plano para esa masa, de manera que

$$\lim_{r \rightarrow \infty} \left(1 - \frac{C \propto M}{r}\right) = 1, \quad \lim_{r \rightarrow \infty} \left(1 - \frac{C \propto M}{r}\right)^{-1} = 1$$

Y la tercera es que en r infinito nuestras ecuaciones se tienen que comportar como en la mecánica clásica (límite newtoniano), es decir, impondremos a las ecuaciones que hemos derivado esas condiciones.

Si partimos de la ecuación geodésica para la componente radial derivada de la solución propuesta particularizada para r, según sus símbolos de Christoffel no nulos

$$\ddot{r} + \Gamma_{rr}^r \dot{r} \dot{r} + \Gamma_{\theta\theta}^r \dot{\theta} \dot{\theta} + \Gamma_{\phi\phi}^r \dot{\phi} \dot{\phi} + \Gamma_{tt}^r \dot{t} \dot{t} = 0$$

siendo el parámetro afín el tiempo propio τ

En el límite newtoniano las velocidades son mucho menores que las de la luz, con lo que cualquier componente que tenga una derivada espacial es despreciable comparada con las otras componentes dependientes del tiempo. Por otra parte cuando el parámetro afín es el tiempo propio su derivada se puede igualar a 1 en el infinito.

$$\Gamma_{rr}^r \dot{r} \dot{r} + \Gamma_{\theta\theta}^r \dot{\theta} \dot{\theta} + \Gamma_{\phi\phi}^r \dot{\phi} \dot{\phi} \approx 0 \text{ y } \dot{t} \approx 1$$

$$\ddot{r} + \Gamma_{tt}^r \dot{t} \dot{t} \approx 0 \Rightarrow \ddot{r} + \Gamma_{tt}^r \approx 0$$

Ahora traemos la expresión de Γ_{tt}^r a partir de la métrica,

$$\Gamma_{tt}^r = \frac{1}{2} g^{rr} (\partial_t g_{rt} + \partial_t g_{tr} - \partial_r g_{tt})$$

Dado que estamos trabajando en una solución estática, cuya métrica no depende del tiempo, todas las derivadas parciales con respecto al tiempo desaparecen

$$\Gamma_{tt}^r = \frac{1}{2} g^{rr} (-\partial_r g_{tt})$$

Y como tenemos que $g^{rr} = 1/g_{rr}$

$$\Gamma_{tt}^r = -\frac{1}{2g_{rr}} (\partial_r g_{tt})$$

Y recordamos que

$$g_{tt} = -c^2 \left(1 - \frac{C}{r}\right) \text{ y } g_{rr} = \left(1 - \frac{C}{r}\right)^{-1}$$

$$\partial_r g_{tt} = \partial_r \left[-\left(1 - \frac{C}{r}\right)\right] = c^2 \frac{C}{r^2}$$

Con lo que

$$\Gamma_{tt}^r = -\frac{1}{2g_{rr}}(\partial_r g_{tt}) = -\frac{1}{2g_{rr}}c^2 \frac{C}{r^2}$$

$$= -c^2 \frac{C}{2r^2} \left(1 - \frac{C}{r}\right)$$

Volviendo a la ecuación geodésica para el radio,

$$\ddot{r} - \frac{C}{2r^2}c^2 \left(1 - \frac{C}{r}\right) \approx 0 \text{ y recordando el límite del campo débil } \lim_{r \rightarrow \infty} \left(1 - \frac{C}{r}\right) = 1 \text{ tenemos que}$$

$$\ddot{r} - \frac{C}{2r^2}[c^2] \approx 0 \Rightarrow \ddot{r} \approx \frac{C}{2r^2}[c^2]$$

Comparando con la aceleración newtoniana, $a = \frac{GM}{r^2}$, en el límite cuando r tiende a infinito $\ddot{r} = a \Rightarrow \frac{C}{2r^2}[c^2] = \frac{GM}{r^2}$ con lo que $C = \frac{2GM}{c^2}$ que en adelante llamaremos radio de Schwarzschild y que denotaremos como $r_s = \frac{2GM}{c^2}$

Y sustituyendo en la métrica,

$$ds^2 = -c^2 \left(1 - \frac{r_s}{r}\right) dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + r^2 d\Omega^2 \text{ donde } r_s = \frac{2GM}{c^2}$$

Que es la forma común de mostrar la métrica de Schwarzschild.

4.4.5. Evaluaciones útiles sobre el radio de Schwarzschild

Trabajando con la expresión anterior para r_s encontramos que, introduciendo ya órdenes de magnitud para las ctes fundamentales

$$r_s = \frac{2GM}{c^2} = \frac{2G}{c^2} M \sim \frac{10^{-11}}{(10^8)^2} M \sim \frac{10^{-11}}{10^{16}}$$

$$\sim 10^{-27} M$$

Y por tanto en órdenes de magnitud

$$r_s \sim 10^{-27} M$$

Llamaremos a la relación entre r_s y M radio de Schwarzschild por unidad de masa y lo denotaremos como r_m . Vemos así que $r_m \sim 10^{-27}$ para cualquier masa M .

Encontramos por tanto que la relación $\frac{r_s}{r}$, que llamaremos compacidad u , es un múltiplo de r_m y por tanto para órdenes de radio n el orden de la compacidad será 10^{-27-n} veces el orden de la masa M .

Es decir, que llamando u a la compacidad

$$u = \frac{r_s}{r} = M \frac{r_m}{r} \Rightarrow r \sim n, M \sim b \text{ y } r_m \sim 10^{-27} \Rightarrow u \sim 10^{-27-n+b}$$

Donde vemos que el orden del radio n y el de la masa M tienen que ser significativos para alterar la compacidad: sólo masas elevadas y radios pequeños elevarán el orden de magnitud de la compacidad y aceptando una precisión de orden x tendremos que

$$x = b - n - 27 \Rightarrow b - n \geq x + 27 \text{ donde } b \text{ es el orden de la masa y } n \text{ el del radio}$$

O de otra forma, para que la curvatura del espacio sea significativa como veremos a continuación.

Siendo la expresión de la métrica

$$ds^2 = -c^2 \left(1 - \frac{r_s}{r}\right) dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + r^2 d\Omega^2$$

Escrita ahora en términos de compacidad

$$ds^2 = -c^2(1 - u)dt^2 + (1 - u)^{-1}dr^2 + r^2 d\Omega^2$$

Vemos que para cualquier coordenada angular, en cualquier radio de la simetría esférica, el orden de la compacidad $\sim 10^{-27-n+b}$ podría hacer que fuera despreciada y por tanto

$$ds^2 = -c^2 dt^2 + dr^2 + r^2 d\Omega^2$$

Que es exactamente la métrica de Minkowski.

La diferencia de órdenes de magnitud $(b - n)$ demuestra ser es un buen criterio para despreciar la curvatura y hablar de campos débiles donde la gravity de Newton, más sencilla, es completamente válida. A esa diferencia se le llama régimen de campo débil.

Como ejemplo para la Tierra $u \sim 10^{-9}$ imponiendo orden 6 al radio y 24 a la masa.

4.4.6. Determinación de geodésicas

Ya podemos determinar las geodésicas en un campo gravitatorio radial recuperando las ecuaciones para las cuatro dimensiones y sustituyendo la cte de integración hallada

$$\ddot{t} + 2f'_t(r) \dot{r} \dot{t} = 0$$

$$\ddot{r} + f'_r(r) \dot{r}^2 - r(e^{-2f_r(r)})(\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2) + c^2 f'_t(r) e^{2f_t(r)-2f_r(r)} \dot{t}^2 = 0$$

$$\ddot{\theta} + (2/r)\dot{r}\dot{\theta} - (1/2)\sin 2\theta \dot{\phi}^2 = 0$$

$$\ddot{\phi} + 2\cot \theta \dot{\theta}\dot{\phi} + 2(1/r)\dot{r}\dot{\phi} = 0$$

Y resumiendo las funciones y sus derivadas

$$e^{2f_t(r)} = 1 - \frac{r_s}{r}$$

$$e^{2f_r(r)} = \frac{1}{1 - \frac{r_s}{r}}$$

$$f'_t(r) = \frac{r_s}{2r(r - r_s)}$$

$$f'_r(r) = \frac{-r_s}{2r(r - r_s)}$$

Tenemos así que las geodésicas resultan

Geodésicas en Schwarzschild	
t	$\ddot{t} + \frac{r_s}{r(r - r_s)} \dot{r} \dot{t} = 0$
r	$\ddot{r} - \frac{r_s}{2r(r - r_s)} \dot{r}^2 - r\left(1 - \frac{r_s}{r}\right)(\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2) + c^2 \frac{r_s}{2r(r - r_s)} \dot{t}^2 = 0$
θ	$\ddot{\theta} + (2/r)\dot{r}\dot{\theta} - (1/2)\sin 2\theta \dot{\phi}^2 = 0$
φ	$\ddot{\phi} + 2\cot \theta \dot{\theta}\dot{\phi} + 2(1/r)\dot{r}\dot{\phi} = 0$

Recordando además que en la métrica se observan las siguientes coordenadas cíclicas o magnitudes conservadas donde ya se puede sustituir la cte de integración

Energía específica $\epsilon = e^{2f_t(r)}\dot{t} = \left(1 - \frac{r_s}{r}\right)\dot{t} \Leftrightarrow \frac{\partial L}{\partial t} = 0 \Leftrightarrow \frac{\partial L}{\partial t} = \epsilon$

Momento angular $j = r^2 \sin^2 \theta \dot{\phi} \Leftrightarrow \frac{\partial L}{\partial \phi} = 0 \Leftrightarrow \frac{\partial L}{\partial \phi} = j$

4.4.7. Aplicación extendida de la solución de Schwarzschild (teorema de Birkhoff)

El teorema de Birkhoff establece que toda solución de las ECE en el vacío con simetría esférica es necesariamente estática y asintóticamente plana (su compacidad u

tiende a 0 para $r \gg r_s$), por lo que coincide con la métrica de Schwarzschild.

Por tanto la solución de Schwarzschild demostrada anteriormente sigue siendo válida y equivalente a la que obtendríamos de introducir el tiempo como variable en las funciones de las coordenadas de la métrica.

También nos permite afirmar que aunque el radio en el que se concentre la masa varíe en el tiempo, siempre que se mantenga la simetría esférica, la solución permanecerá constante, o que considerando que la simetría se mantiene, la solución no variará ante cambios en la densidad interna de, por ejemplo, el sol y los planetas del sistema solar incluida la Tierra.

4.4.8. Observadores locales (shell)

Ahora vamos a observar que los móviles nunca perciben un cambio de posición con respecto a sí mismos, sólo ven pasar su tiempo propio. Si además siguen una trayectoria geodésica están en caída libre en el campo radial que la genera con lo que por el principio de equivalencia sólo perciben una métrica plana de la forma

$$ds^2 = -c^2 d\tau^2$$

Pero los sistemas de referencia que no experimenten esa caída libre sí perciben ese movimiento, que además tiene que satisfacer la invarianza de ds^2 para cualquier observador, con lo que

$$ds_{cayendo}^2 = ds_{no cayendo}^2$$

es decir,

$$-c^2 d\tau_{cayendo}^2 = -c^2 \left(1 - \frac{r_s}{r}\right) dt_{no cayendo}^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr_{nocayendo}^2 + r^2 d\Omega_{nocayendo}^2$$

que es la ecuación que nos permite deducir velocities cercanas al observador en caída y que llamamos observaciones locales u observaciones shell, por considerar que se producen en una de las superficies esféricas de r constante de la métrica cerca del fenómeno que se quiere caracterizar.

Dichas velocities no son las mismas que se pueden obtener dividiendo la métrica, asintóticamente plana, por dt , que toman valores de velocity válidos para observadores coordinados, es decir en el infinito, pero que

no permiten operaciones de adición de geometría plana imposibles con curvatura.

En todo caso la relación entre velocidades de observadores coordinados y observadores locales o shell tienen que seguir la relación de dilatación del tiempo explicada [aquí](#) y que se muestra a continuación

$$\frac{d\tau}{dt} = \sqrt{\left(1 - \frac{r_s}{r}\right)}$$

que como vemos es lo que permite que en sucesos alejados del radio de Schwarzschild podamos usar el tiempo propio del móvil que cae para caracterizar la caída desde otro móvil estático respecto a ella.

4.5. Cambios rígidos de coordenadas aplicados al principio de mínima acción

Vamos a dejar aquí reflejadas algunas reflexiones sobre cómo afectan los cambios de coordenadas rígidos al principio de mínima acción.

Para el lagrangiano $L = \left(\frac{ds}{d\lambda}\right)^2$ y la métrica $ds^2 = g_{ab}dx^a dx^b$ expresada en notación tensorial, se elige un parámetro afín λ en el que L se mantiene cte $\frac{\partial L}{\partial \lambda} = 0$.

Se utiliza la notación $\dot{x} = \frac{dx}{d\lambda}$ con respecto a ese parámetro afín y se define el cambio de variable $x^a = x^a_0 + \alpha^a$ donde α^a es una constante con respecto al parámetro afín (condición 1) y al lagrangiano (condición 2).

Resulta así que $\dot{x}^a = \dot{x}^a_0$ y $\ddot{x}^a = \ddot{x}^a_0$ porque α^a desaparece al derivar (condición 1) y que podemos seguir formulando las ecuaciones de Euler-Lagrange con este lagrangiano (condición 2), aunque no podremos introducirlo en las expresiones tensoriales genéricas porque estamos estudiando el caso de un único cambio de variable.

Para cada variable x^a tenemos que la mínima acción por Euler-Lagrange es de la forma

$$(Ec1) \frac{d}{d\lambda} \left(\frac{\partial L}{\partial \dot{x}^a} \right) - \frac{\partial L}{\partial x^a} = 0$$

y tenemos que desarrollar

$$T1 = \left(\frac{\partial L}{\partial \dot{x}^a} \right), T2 = \frac{d}{d\lambda} \left(\frac{\partial L}{\partial \dot{x}^a} \right) \text{ y } T3 = \left(\frac{\partial L}{\partial x^a} \right)$$

Desarrollo de T1

$$(T1) \frac{\partial L}{\partial \dot{x}^a} = \frac{\partial}{\partial \dot{x}^a} \left(\frac{ds}{d\lambda} \right)^2 = \frac{\partial}{\partial \dot{x}^a} \left(\frac{1}{d\lambda^2} g_{ab} dx^a dx^b \right)$$

Considerando que todas las variables del lagrangiano son independientes esta expresión sólo tomará un valor distinto de cero para $\dot{x}^a$ y sus términos cruzados

$$\frac{\partial L}{\partial \dot{x}^a} = 2g_{ab}\dot{x}^b$$

Recordando que no podemos utilizar $\dot{x}^a = \dot{x}^a_0$ de la definición del cambio de coordenadas porque sólo es válido en el caso de que a sea igual a b en la expresión, por lo que

$$\frac{\partial L}{\partial \dot{x}^a} = 2g_{ab}\dot{x}^b \neq 2g_{ab}\dot{x}^b_0$$

Es decir, que un cambio de coordenadas en x^a afectará al término $\frac{\partial L}{\partial \dot{x}^a}$ sólo en $\dot{x}^a$ y sus términos cruzados (cuando b tome el valor a en la expresión anterior).

Ejemplos

En Minkowski en coordenadas esféricas con $ds^2 = -c^2 dt^2 + dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2$ el término $\frac{\partial L}{\partial \dot{x}^a}$ sólo variará ante cambios de coordenadas en r y θ pero no en ϕ ni en t .

En Minkowski en coordenadas cartesianas $ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2$ el término $\frac{\partial L}{\partial \dot{x}^a}$ no variará ante cambios de coordenadas en ninguna variable.

En Schwarzschild con $ds^2 = -A(r)c^2 dt^2 + B(r)dr^2 + C(r)d\theta^2 + D(r,\theta)d\phi^2$ el término $\frac{\partial L}{\partial \dot{x}^a}$ variará ante cambios de coordenadas en r pero no en t , θ ni ϕ .

Desarrollo de T2

$$(T2) \frac{d}{d\lambda} \left(\frac{\partial L}{\partial \dot{x}^a} \right) = \frac{d}{d\lambda} (2g_{ab}\dot{x}^b)$$

Considerando que ahora podemos encontrar términos cruzados donde dos variables dependan del parámetro tendremos que realizar la regla del producto de la forma

$$\begin{aligned}\frac{d}{d\lambda}\left(\frac{\partial L}{\partial \dot{x}^a}\right) &= \frac{d}{d\lambda}(2g_{ab}\dot{x}^b) = 2\frac{d}{d\lambda}(g_{ab}\dot{x}^b) = \\ &2\left(\frac{dg_{ab}}{d\lambda}\dot{x}^b + g_{ab}\frac{d\dot{x}^b}{d\lambda}\right) = 2\left(\frac{dg_{ab}}{d\lambda}\dot{x}^b + g_{ab}\ddot{x}^b\right) \\ &= 2\frac{dg_{ab}}{d\lambda}\dot{x}^b + 2g_{ab}\ddot{x}^b\end{aligned}$$

Descomponemos la derivada de la métrica introduciendo un índice diferente μ

$$\frac{dg_{ab}}{d\lambda} = \frac{\partial g_{ab}}{\partial x^\mu} \frac{dx^\mu}{d\lambda} = \frac{\partial g_{ab}}{\partial x^\mu} \dot{x}^\mu = (\partial_\mu g_{ab})\dot{x}^\mu$$

Por tanto volviendo a la expresión anterior

$$\begin{aligned}\frac{d}{d\lambda}\left(\frac{\partial L}{\partial \dot{x}^a}\right) &= 2\frac{dg_{ab}}{d\lambda}\dot{x}^b + 2g_{ab}\ddot{x}^b \\ &= 2(\partial_\mu g_{ab})\dot{x}^\mu \dot{x}^b + 2g_{ab}\ddot{x}^b\end{aligned}$$

Recordando cómo se define el cambio de coordenadas tenemos que analizar cómo se altera la expresión anterior cuando los índices mudos toman el valor a , momento en el que $\dot{x}^a = \dot{x}^a_0$ y $\ddot{x}^a = \ddot{x}^a_0$

$$\frac{d}{d\lambda}\left(\frac{\partial L}{\partial \dot{x}^a}\right) = 2(\partial_\mu g_{ab})\dot{x}^\mu \dot{x}^b + 2g_{ab}\ddot{x}^b$$

El término $g_{ab}\ddot{x}^b$ sólo se alterará cuando $a = b$, donde será sensible al cambio de coordenadas en x^a , con lo que podemos escribir que $2g_{ab}\ddot{x}^b \neq 2g_{ab}\ddot{x}^b_0$

El término $(\partial_\mu g_{ab})\dot{x}^\mu \dot{x}^b$ sólo se alterará cuando $a = b$ con independencia de que $a = \mu$, pues $\dot{x}^a$ aparecerá como factor común del sumatorio de las derivadas parciales de μ . Así podemos escribir que $(\partial_\mu g_{ab})\dot{x}^\mu \dot{x}^b = 2(\partial_\mu g_{ab})\dot{x}^\mu \dot{x}^b_0$ y observar además que no violamos ninguna regla explícita en la sintaxis del cálculo tensorial (se respetan los índices mudos y libres).

Es decir, que un cambio de coordenadas en x^a afectará al término $\frac{d}{d\lambda}\left(\frac{\partial L}{\partial \dot{x}^a}\right)$ sólo en $\dot{x}^a$ y sus términos cruzados (cuando b tome el valor a en la expresión anterior), al igual que le sucedía al término $\frac{\partial L}{\partial \dot{x}^a}$ antes de hacer la derivada total con respecto al parámetro.

Ejemplos

En Minkowski en coordenadas esféricas con $ds^2 = -c^2 dt^2 + dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2$ el término $\frac{d}{d\lambda}\left(\frac{\partial L}{\partial \dot{x}^a}\right)$ sólo variará ante cambios de coordenadas en r y θ pero no en ϕ ni en t .

En Minkowski en coordenadas cartesianas $ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2$ el término $\frac{d}{d\lambda}\left(\frac{\partial L}{\partial \dot{x}^a}\right)$ no variará ante cambios de coordenadas en ninguna variable.

En Schwarzschild con $ds^2 = -A(r)c^2 dt^2 + B(r)dr^2 + C(r)d\theta^2 + D(r,\theta)d\phi^2$ el término $\frac{d}{d\lambda}\left(\frac{\partial L}{\partial \dot{x}^a}\right)$ variará ante cambios de coordenadas en r pero no en t , θ ni ϕ .

Desarrollo de T3

$$(T3) \frac{\partial L}{\partial x^a} = \frac{\partial}{\partial x^a} \left(\frac{ds}{d\lambda}\right)^2 = \frac{\partial}{\partial x^a} \left(\frac{1}{d\lambda^2} g_{ab} dx^a dx^b\right)$$

Aquí se observa que considerando todas las variables independientes la expresión tomará un valor distinto de cero en x^a sólo para sus términos cruzados con la forma

$$\frac{\partial L}{\partial x^a} = \frac{\partial}{\partial x^a} \left(\frac{ds}{d\lambda}\right)^2 = \frac{\partial}{\partial x^a} \left(\frac{1}{d\lambda^2} g_{ab} dx^a dx^b\right) = (\partial_a g_{b\mu}) \dot{x}^b \dot{x}^\mu$$

Que se verá siempre alterada para cualquier cambio de coordenadas en x^a siempre que aparezca en un término cruzado y se transformará según la definición $x^a = x^a_0 + \alpha^a$ donde α^a aparecerá explícitamente por no ser eliminada en ninguna derivación.

Si el lagrangiano es independiente de la variable x^a su derivada en la variable será nula y por tanto el lagrangiano será cte en esa derivada, es decir, que si $\left(\frac{\partial L}{\partial x^a}\right) = 0$ entonces $\frac{d}{d\lambda}\left(\frac{\partial L}{\partial \dot{x}^a}\right) = 0$ y por tanto $\frac{\partial L}{\partial \dot{x}^a} = cte$. Eso significará, que es la aplicación directa del teorema de Noether, que nos encontraremos en el caso de una magnitud conservada, $\frac{\partial L}{\partial \dot{x}^a}$. Y por tanto podemos concluir que un cambio de coordenadas en x^a no alterará la conservación de la magnitud asociada a esa variable, dado que $\dot{x}^a = \dot{x}^a_0$ y por tanto

$$\frac{\partial L}{\partial \dot{x}^a} = \frac{\partial L}{\partial \dot{x}^a_0} = cte$$

Ejemplos

En Minkowski en coordenadas esféricas con $ds^2 = -c^2 dt^2 + dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2$ el término $\frac{\partial L}{\partial x^a}$ sólo variará ante cambios de coordenadas en r y θ pero no en ϕ ni en t . Las magnitudes conservadas en t y ϕ seguirán conservándose ante cambios de coordenadas en sí mismas, pero no en cambios en r y θ .

En Minkowski en coordenadas cartesianas $ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2$ el término $\frac{\partial L}{\partial x^a}$ no variará ante cambios de coordenadas en ninguna variable. Las magnitudes conservadas en las cuatro variables seguirán conservándose ante cualquier cambio de coordenadas.

En Schwarzschild con $ds^2 = -A(r)c^2 dt^2 + B(r)dr^2 + C(r)d\theta^2 + D(r,\theta)d\phi^2$ el término $\frac{\partial L}{\partial x^a}$ variará ante cambios de coordenadas en r pero no en t , θ ni ϕ . Las magnitudes conservadas en t y ϕ seguirán conservándose ante cambios de coordenadas en sí mismas, pero no en cambios en r y θ .

Resumen

Vamos a integrar los términos anteriores en la expresión de la geodésica Ec1 teniendo cuidado de no provocar duplicidad de índices que los hagan mudos. Así tenemos

$$(Ec1) \frac{d}{d\lambda} \left(\frac{\partial L}{\partial \dot{x}^a} \right) - \frac{\partial L}{\partial x^a} = 0$$

$$\text{Que con } T2 = \frac{d}{d\lambda} \left(\frac{\partial L}{\partial \dot{x}^a} \right) \text{ y } T3 = \left(\frac{\partial L}{\partial x^a} \right) \text{ resulta}$$

$$T2 - T3 = 0$$

Comprobamos antes que

$$T2 = 2 (\partial_\mu g_{ab}) \dot{x}^\mu \dot{x}^b + 2 g_{ab} \ddot{x}^b$$

$$T3 = (\partial_a g_{b\mu}) \dot{x}^b \dot{x}^\mu$$

Pero hay que alterar los índices en $T2$ y $T3$ para evitar que violen la sintaxis tensorial cuando restemos $T2$ y $T3$ (el índice b aparecería tres veces en el sumatorio).

$$\text{Podemos escribir así } 2 (\partial_\mu g_{ab}) = (\partial_\mu g_{ab} + \partial_b g_{a\mu})$$

$$\text{Para reescribir } T2 \text{ como } T2 = (\partial_\mu g_{ab} + \partial_b g_{a\mu}) \dot{x}^\mu \dot{x}^b + 2 g_{ab} \ddot{x}^b$$

Con lo que Ec1 resulta

$$(\partial_\mu g_{ab} + \partial_b g_{a\mu}) \dot{x}^\mu \dot{x}^b + 2 g_{ab} \ddot{x}^b - (\partial_a g_{b\mu}) \dot{x}^b \dot{x}^\mu = 0$$

Siendo la ecuación canónica de la geodésica según el principio de mínima acción para la coordenada x^a

$$(Ec2) \frac{d^2 x^a}{d\lambda^2} + \Gamma_{bc}^a \frac{dx^b}{d\lambda} \frac{dx^c}{d\lambda} = \ddot{x}^a + \Gamma_{bc}^a \dot{x}^b \dot{x}^c = 0$$

Queremos integrar las conclusiones anteriores en Ec1 para derivar cómo se ve alterada ante el cambio de variable propuesto en x^a y cómo se alteran sus christoffels Γ_{bc}^a en particular.

Alterando el orden de los términos en la igual Ec1

$$2 g_{ab} \ddot{x}^b + (\partial_\mu g_{ab} + \partial_b g_{a\mu}) \dot{x}^\mu \dot{x}^b - (\partial_a g_{b\mu}) \dot{x}^b \dot{x}^\mu = 0$$

Sacando en el segundo término $\dot{x}^b \dot{x}^\mu$ como factor común

$$2 g_{ab} \ddot{x}^b + (\partial_\mu g_{ab} + \partial_b g_{a\mu} - \partial_a g_{b\mu}) \dot{x}^b \dot{x}^\mu = 0$$

Y buscando asemejarla a la expresión canónica es importante tener en cuenta que

$$g^{ad} g_{ab} = \delta_b^d$$

Así que vamos a multiplicar toda la expresión por un factor $(1/2)$ y la inversa de la métrica g^{ad} (evitamos introducir g^{ab} porque queremos evitar duplicar el índice b y c ya lo hemos usado para una coordenada en la geodésica) resultando

$$\delta_b^d \ddot{x}^b + (1/2) g^{ad} (\partial_\mu g_{ab} + \partial_b g_{a\mu} - \partial_a g_{b\mu}) \dot{x}^b \dot{x}^\mu = 0$$

Y como $\delta_b^d \ddot{x}^b$ colapsa el índice b a d tenemos que $\delta_b^d \ddot{x}^b = \ddot{x}^d$ y la expresión resulta

$$\ddot{x}^d + (1/2) g^{ad} (\partial_\mu g_{ab} + \partial_b g_{a\mu} - \partial_a g_{b\mu}) \dot{x}^b \dot{x}^\mu = 0$$

Y volviendo a la comparación con la ecuación canónica

$$(Ec2) \ddot{x}^a + \Gamma_{bc}^a \dot{x}^b \dot{x}^c = 0$$

obtenemos la expresión analítica para los christoffels

$$(Ec3) \quad \Gamma_{b\mu}^d = (1/2)g^{ad}(\partial_\mu g_{ab} + \partial_b g_{a\mu} - \partial_a g_{b\mu})$$

Con lo que vemos que en la ecuación (Ec2) un cambio rígido de coordenadas en la variable x^a altera la expresión de la métrica g^{ab} en la ecuación 3 (Ec3) y la de los christoffels Γ^a y como la ecuación geodésica es covariante ante cambios de coordenadas las expresiones de $\dot{x}^b$ y $\dot{x}^c$ tienen que variar con el christoffel para mantener la igualdad.

4.6. Ecuación geodésica canónica según el principio de mínima acción

Aquí vamos a demostrar la forma canónica de la ecuación geodésica según el principio de mínima acción que se propone como un buen ejercicio para trabajar con notación tensorial, definiendo el lagrangiano

$$L = \frac{ds^2}{d\lambda^2} = \frac{g_{\mu\nu} dx^\mu dx^\nu}{d\lambda d\lambda} = g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu \text{ con notación } \frac{dx^a}{d\lambda} = \dot{x}^a \text{ y parámetro afín } \frac{dL}{d\lambda} = 0$$

Formulación de mínima acción Euler-Lagrange

$$\frac{d}{d\lambda} \frac{\partial L}{\partial \dot{x}^\sigma} - \frac{\partial L}{\partial x^\sigma} = 0$$

Cálculo de $\frac{\partial L}{\partial \dot{x}^\sigma}$

$$\begin{aligned} \frac{\partial L}{\partial \dot{x}^\sigma} &= \frac{\partial}{\partial \dot{x}^\sigma} g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu \\ &= g_{\mu\nu} \dot{x}^\nu \frac{\partial}{\partial \dot{x}^\sigma} \dot{x}^\mu \\ &\quad + g_{\mu\nu} \dot{x}^\mu \frac{\partial}{\partial \dot{x}^\sigma} \dot{x}^\nu \end{aligned}$$

y recordando que en notación tensorial $\frac{\partial \dot{x}^a}{\partial \dot{x}^b} = \delta_b^a$

$$\begin{aligned} \frac{\partial L}{\partial \dot{x}^\sigma} &= g_{\mu\nu} \dot{x}^\nu \delta_\sigma^\mu + g_{\mu\nu} \dot{x}^\mu \delta_\sigma^\nu \\ &= g_{\mu\nu} [\delta_\sigma^\mu \dot{x}^\nu + \delta_\sigma^\nu \dot{x}^\mu] \\ &= g_{\sigma\nu} \dot{x}^\nu + g_{\mu\sigma} \dot{x}^\mu = 2g_{\mu\sigma} \dot{x}^\mu \end{aligned}$$

Cálculo de $\frac{d}{d\lambda} \frac{\partial L}{\partial \dot{x}^\sigma}$

$$\begin{aligned} \frac{d}{d\lambda} \frac{\partial L}{\partial \dot{x}^\sigma} &= \frac{d}{d\lambda} 2g_{\mu\sigma} \dot{x}^\mu = 2 \frac{d}{d\lambda} g_{\mu\sigma} \dot{x}^\mu \\ &= 2\dot{x}^\mu \frac{d}{d\lambda} g_{\mu\sigma} + 2g_{\mu\sigma} \ddot{x}^\mu \end{aligned}$$

$\frac{d}{d\lambda} g_{\sigma\mu}$ se puede expresar como $\partial_a g_{\sigma\mu} \dot{x}^a$ y por tanto

$$\frac{d}{d\lambda} \frac{\partial L}{\partial \dot{x}^\sigma} = 2\dot{x}^\mu \partial_\nu g_{\sigma\mu} \dot{x}^\nu + 2g_{\sigma\mu} \ddot{x}^\mu$$

Cálculo de $\frac{\partial L}{\partial x^\sigma}$

$$\begin{aligned} \frac{\partial L}{\partial x^\sigma} &= \frac{\partial}{\partial x^\sigma} g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu = \dot{x}^\mu \dot{x}^\nu \frac{\partial}{\partial x^\sigma} g_{\mu\nu} \\ &= \dot{x}^\mu \dot{x}^\nu \partial_\sigma g_{\mu\nu} \end{aligned}$$

Vuelta a Euler-Lagrange

$$\begin{aligned} \frac{d}{d\lambda} \frac{\partial L}{\partial \dot{x}^\sigma} - \frac{\partial L}{\partial x^\sigma} &= 0 \Rightarrow \frac{d}{d\lambda} \frac{\partial L}{\partial \dot{x}^\sigma} - \frac{\partial L}{\partial x^\sigma} = 0 \\ 2\dot{x}^\mu \partial_\nu g_{\sigma\mu} \dot{x}^\nu + 2g_{\sigma\mu} \ddot{x}^\mu - \dot{x}^\mu \dot{x}^\nu \partial_\sigma g_{\mu\nu} &= 0 \end{aligned}$$

Dividiendo por 2

$$\dot{x}^\mu \partial_\nu g_{\sigma\mu} \dot{x}^\nu + g_{\sigma\mu} \ddot{x}^\mu - \frac{1}{2} \dot{x}^\mu \dot{x}^\nu \partial_\sigma g_{\mu\nu} = 0$$

Por simetría $\partial_\nu g_{\sigma\mu} \dot{x}^\nu \dot{x}^\mu$ se puede escribir como $\frac{1}{2} \partial_\mu g_{\nu\sigma} \dot{x}^\mu \dot{x}^\nu + \frac{1}{2} \partial_\nu g_{\mu\sigma} \dot{x}^\mu \dot{x}^\nu$ y por tanto

$$\begin{aligned} \frac{1}{2} \partial_\mu g_{\nu\sigma} \dot{x}^\mu \dot{x}^\nu + \frac{1}{2} \partial_\nu g_{\mu\sigma} \dot{x}^\mu \dot{x}^\nu + g_{\sigma\mu} \ddot{x}^\mu \\ - \frac{1}{2} \dot{x}^\mu \dot{x}^\nu \partial_\sigma g_{\mu\nu} &= 0 \\ \frac{1}{2} \partial_\mu g_{\nu\sigma} \dot{x}^\mu \dot{x}^\nu + \frac{1}{2} \partial_\nu g_{\mu\sigma} \dot{x}^\mu \dot{x}^\nu - \frac{1}{2} \dot{x}^\mu \dot{x}^\nu \partial_\sigma g_{\mu\nu} \\ + g_{\sigma\mu} \ddot{x}^\mu &= 0 \\ \frac{1}{2} (\partial_\mu g_{\nu\sigma} \dot{x}^\mu \dot{x}^\nu + \partial_\nu g_{\mu\sigma} \dot{x}^\mu \dot{x}^\nu - \dot{x}^\mu \dot{x}^\nu \partial_\sigma g_{\mu\nu}) \\ + g_{\sigma\mu} \ddot{x}^\mu &= 0 \end{aligned}$$

$$\frac{1}{2} \dot{x}^\mu \dot{x}^\nu (\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu}) + g_{\sigma\mu} \ddot{x}^\mu = 0$$

Formulación analítica de los símbolos de Christoffel

$$\Gamma_{\mu\nu}^\lambda = \frac{1}{2} g^{\lambda\sigma} (\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu})$$

Metiendo los Christoffel en Euler-Lagrange

$$\begin{aligned} \frac{1}{2} \dot{x}^\mu \dot{x}^\nu (\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu}) + g_{\sigma\mu} \ddot{x}^\mu &= 0 \\ g_{\sigma\mu} \ddot{x}^\mu + g_{\sigma\lambda} \Gamma_{\mu\nu}^\lambda \dot{x}^\mu \dot{x}^\nu &= 0 \end{aligned}$$

Multiplicando por $g^{\rho\sigma}$

$$g^{\rho\sigma} g_{\sigma\mu} \ddot{x}^\mu + g^{\rho\sigma} g_{\sigma\lambda} \Gamma_{\mu\nu}^\lambda \dot{x}^\mu \dot{x}^\nu = 0$$

$$\delta_\mu^\rho \ddot{x}^\mu + \delta_\lambda^\rho \Gamma_{\mu\nu}^\lambda \dot{x}^\mu \dot{x}^\nu = 0$$

$$\ddot{x}^\rho + \Gamma_{\mu\nu}^\rho \dot{x}^\mu \dot{x}^\nu = 0$$

Resultado en forma canónica

$$\ddot{x}^a + \Gamma_{bc}^a \dot{x}^b \dot{x}^c = 0$$

5. Acknowledgements

This document arises as a final course project for a remote training program provided by the Universidad Nacional de Educación a Distancia (UNED) as part of its permanent education study program, "[Geometry of Spacetime. Black Holes and Gravitational Waves](#)," and hence it contains general physics content that exceeds the context of satellite navigation. While the course aimed to explain and apply general relativity in these two cosmological phenomena, I have decided to be humble in my aspirations regarding the syllabus and apply the physics presented to a case where I felt more comfortable and could verify that the conclusions are indeed correct. Thus, I have reserved cosmology to study it without fear of making errors, leaving it for my leisure time.

This course has been my first university-level approach to general relativity and cosmology, and for that, I am extremely grateful to my course professors Carlos Shahbazi Alonso and Pablo Bueno Gómez for their lessons, their initiative in inaugurating and facilitating this course, and all the material they have made available to the students, as well as to my fellow classmates, for being the most recommendable and fortunate company one could hope for in this endeavor.

Should any error be found in the document, it will be the sole responsibility of the author, given that, as can be read, I have exceeded the required proofs that I could have copied directly from more than one classic text or technical manual, but which I have

developed in order to better understand the course contents.

And as a curiosity for those interested readers who are fans of science fiction and probably know the film *Interstellar* (2014) by Christopher Nolan: at one point in the movie, the protagonists have to separate; some land on a planet near a black hole and others remain far from it, and when they meet again, time has passed much slower for those who landed on the planet. They will find a graph that explains it [here](#). I hope you savor this moment as much as I did when I developed this graph and thought that its consequences were exactly those shown in the film, which was advised by none other than Kip Thorne, winner of the Nobel Prize in Physics in 2017 for his contribution to the understanding of gravitational waves.

And to point out any of those errors or any other topic of interest relative to it, I leave my contact here: fer.urd@gmail.com.

Fernando Urdiales
Madrid, mayo de 2026